

IRS-Enhanced Spectrum Sensing and Secure Transmission in Cognitive Radio Networks

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Abstract—Spectrum sensing and communication security are of crucial importance in cognitive radio networks (CRNs). In this paper, we utilize intelligent reflecting surfaces (IRS) to simultaneously enhance spectrum sensing accuracy and the secrecy performance of secondary users (SUs) through physical layer security (PLS) techniques. Additionally, we employ IRS as a novel approach to achieve the target probability of detection. We formulate a joint sensing and transmission security optimization problem to maximize the sum secrecy rate of SUs under both perfect and imperfect channel state information (CSI). To transform the probability of detection into a tractable expression, we adopt a safe approximation for the Q -function. We use a computationally-efficient block coordinate descent (BCD)-based algorithm to optimize the beamforming design and IRS phase shifts alternately. Specifically, we employ the $\mathcal{S}$ -procedure to handle the semi-infinite constraints under the imperfect CSI case. Simulation results demonstrate that by leveraging IRS for spectrum sensing, we can significantly reduce the sensing time while achieving the required probability of detection and the probability of false alarm. Furthermore, our proposed scheme improves both sensing accuracy and secrecy rate in both cases compared to the benchmark schemes.

Index Terms—Intelligent reflecting surface (IRS), secure transmission, cognitive radio, spectrum sensing, physical layer security.

I. INTRODUCTION

IN RECENT years, spectrum resources have become increasingly scarce due to the proliferation of wireless communication devices and the emergence of diverse

ultra-wideband communication services [1]. To enhance spectrum efficiency (SE), cognitive radio (CR) has been proposed for its unique mechanism, allowing secondary users (SUs) to access the spectrum licensed to primary users (PUs) using various operational paradigms [2], [3], [4], which include spectrum sharing, sensing-enhanced spectrum sharing, and opportunistic spectrum access. In the case of opportunistic spectrum access, SUs utilize shared spectrum only when PUs are detected to be inactive, thus improving SE while ensuring the quality of service (QoS) of the primary network [5].

One major factor affecting the performance of an opportunistic spectrum access CRN is the accuracy of spectrum sensing. Thus, several spectrum sensing schemes, such as the energy detection [5], [6], [7], the eigenvalue detection [8] and the matched filtering detection [9] have been developed for improving the sensing performance in different communication scenarios. Energy detection is very effective and has been widely applied in CRNs since it does not need the priori knowledge of the signals of PUs and strict phase synchronization. Besides, to tackle the hidden terminal, shadowing and multipath fading problems, cooperative spectrum sensing [10] has been proposed that multiple SUs can cooperate to perform spectrum sensing. Recently, the techniques of deep transfer [11], unmanned aerial vehicle (UAV) [12], online detection [13] and machine learning [14] have been integrated into cooperative sensing to achieve better sensing performance. Although the existing approaches have made great breakthroughs, the sensing performance is still very limited when the direct link between the PU and the sensing node is blocked or very weak.

Moreover, since the spectrum band in CRNs is shared, the communication security issues become more complicated and can be caused by many conditions, e.g. the PU emulation, jamming and eavesdropping. Eavesdropping exists in the physical layer where the potential eavesdroppers (Eves) can intercept the confidential information that transmitted to the legitimate users, which causes severe damages to the communication security of CRNs. To address this issue, two major secure communication techniques have been proposed. One is the traditional cryptographic techniques and the other is the physical layer security (PLS) [15]. The PLS technique has attracted great attention these years for its computationally-efficiency and high effectiveness in ensuring the secure transmission. To further enhance the PLS, cooperative jamming (CJ) [15], relay-aided secure communications [16] and artificial noise (AN)-aided beamforming [17]. However, employing relays and multiple antennas incurs an extra cost. Moreover, although

Manuscript received 16 December 2022; revised 4 November 2023; accepted 15 February 2024. Date of publication 5 March 2024; date of current version 14 August 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62271267, Grant 62222107, Grant 62071223, and Grant 62031012; in part by the National Research Foundation, Singapore; and in part by the Infocomm Media Development Authority under its Future Communications Research and Development Program. An earlier version of this paper was presented in part at the IEEE International Conference on Communications 2023 [DOI: 10.1109/ICC45041.2023.10279590]. The associate editor coordinating the review of this article and approving it for publication was S. Dey. (Corresponding author: Wei Wu.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TWC.2024.3370812>.

Digital Object Identifier 10.1109/TWC.2024.3370812

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AN and CJ can downgrade the quality of eavesdropping, they inevitably cause interference to the legitimate users and cannot guarantee a satisfactory secrecy performance. Thus, there is an urgent need for new techniques to tackle these problems.

Recently, the intelligent reflecting surface (IRS) technique is envisioned to be a key enabler for the next generation wireless communication networks [18], [19], [20], [21]. In particular, an IRS is a planar array metasurface with a large number of nearly-passive and low-cost reflective elements. Reference [44] The phases of IRS elements can be adjusted to adapt to the real-time channel conditions, thereby able to reshape the end-to-end communication environment, for example, enhancing the signal power gain at the desired receivers, harnessing the multiuser interference and degrading the channel quality of wiretap links [20]. The works in [21], [22], [23], [24], [25], and [26] have shown that IRS can greatly improve the spectrum and energy efficiency and enhance the robustness in CRNs. Due to the above advantages, it is envisioned that IRS can be exploited to improve the spectrum sensing accuracy and enhance the PLS in opportunistic access CRNs. A joint sensing and transmission security optimization problem is investigated to further improve the secrecy performance of IRS-assisted CRNs. The related works and the contribution of our work are summarized as follows.

A. Related Works

The related works have extensively investigated secure CRNs based on PLS. For example, in [27], the secrecy rate maximization for secure multiple-input single-output (MISO) CRNs was first studied under the full channel state information (CSI), the transmit covariance matrix was optimized under the joint transmit power and interference power constraints. Based on the work in [27], the authors in [28] considered the imperfect CSI of the secondary, primary and eavesdropper channels in secure MISO CRNs. In [29], the secure multi-input multi-output (MIMO) CRNs with simultaneous wireless information and power transfer (SWIPT) was studied, where the secrecy rate was maximized by optimizing the precoding matrix and the AN. In [30] and [31], the PLS design for CRNs with non-orthogonal multiple access (NOMA) was investigated, where the performance metrics analysis and the optimization of the CJ-aided beamforming design were respectively presented.

For IRS-enhanced spectrum sensing, the authors in [32] first proposed an IRS-enhanced energy detection scheme for the single-user spectrum sensing, cooperative sensing and diversity reception in CRNs. Results in [32] demonstrated that the probability of detection can be significantly improved by the exploitation of IRS. Then, the authors in [33] exploited IRS to improve the performance of crowd spectrum sensing (CSS) for Internet of things (IoT) networks, in which the closed-form expressions of the crowd detection performance metrics were derived. In [34], an IRS-assisted weighted energy detection (WED) scheme for spectrum sensing was proposed, where the weighted coefficients were optimized to minimize the probability of miss detection at a given probability of false alarm. The authors in [35] incorporated

the statistical CSI, the IRS phase shifts optimization and the location information into IRS-enhanced spectrum sensing. Then, the authors in [36] considered a joint sensing and transmission optimization for IRS-assisted CRNs, where the sensing time, the transmit beamforming and the IRS phase shifts for the spectrum sensing and the secondary transmission were respectively optimized to facilitate spectrum sensing and enhance the secondary transmission rate. In [37], the authors derived the number of IRS reflecting elements required to achieve the probability of detection that close to 1.

On the other hand, the combination of IRS and PLS has been investigated in the existing literatures. The key idea is to enhance the signals received at the legitimate users and suppress the signals at the Eves by adjusting the IRS phase shifts. The original design for IRS-assisted secure communication networks was studied in [38] under MISO settings with a multi-antenna base station (BS), one single-antenna legitimate user and one single-antenna Eve, the secrecy rate was maximized by optimizing the transmit beamforming at the BS and the phase shifts of IRS under the maximum power and the IRS phase gain constraints. Results in [38] have shown that IRS can greatly improve the system secrecy rate. The authors in [39] considered the imperfect CSI of wiretap channels in an IRS-assisted secure MISO system with multiple legitimate users, Eves and IRSs. The transmit beamforming at the BS, AN covariance and IRS phase shifts were jointly optimized for the maximization of the achievable rate.

The authors in [40] studied the energy efficient design for IRS-assisted MISO networks, where the energy efficiency was maximized by jointly designing the transmit and jamming beamforming and IRS phase shifts under both the perfect and imperfect CSI. Since the imperfect CSI is considered, one major difficulty is the infinite inequality constraints caused by CSI uncertainties. Thus, the $\mathcal{S}$ -procedure was leveraged in [39] and [40] to transform the infinite inequality constraints into tractable linear matrix inequalities (LMIs). Moreover, the authors in [41] studied an IRS-assisted secure MIMO networks with AN where the secrecy rate was maximized by jointly optimizing the transmit covariance, AN covariance and the IRS phase shifts.

Especially, the works in [42], [43], [44], and [45] have shown that IRS is applicable for the PLS design in CR scenarios. In particular, the authors in [42] proposed an IRS-assisted secure spectrum sharing-based CRNs, where the secrecy rate was maximized under three conditions of Eve CSI: full CSI, imperfect CSI and no CSI. Else, the authors in [43] studied an IRS-assisted secure MIMO CRNs, where the secrecy rate was maximized by jointly optimizing the transmit covariance at the BS and the IRS phase shifts. The energy efficient design for IRS-assisted secure CRNs was proposed in [45], where the trade off between the secrecy rate and energy consumption was established to maximize the secrecy energy efficiency (SEE).

B. Contribution and Organization

Although the conventional and IRS-assisted secure CRNs based on PLS have been studied in [27], [28], [29], [30],

and [31] and [42], [43], [44], and [45], respectively. They mainly focus on the spectrum sharing paradigm and IRS is exploited only to assist the secure transmission in [42], [43], [44], and [45]. However, to our best knowledge, there is no study that has considered the PLS design for IRS-assisted secure CRNs under the opportunistic access paradigm. Moreover, the works in [32], [34], [35], [36], and [37] only considered exploiting IRS to assist spectrum sensing but not the secure transmission.

To date, security considerations have not been a focal point in IRS-assisted CRNs operating under the opportunistic spectrum access mode. Additionally, there is limited research on the joint design of IRS-enhanced spectrum sensing and secure transmission. Furthermore, existing literature lacks investigations into designing an appropriate IRS-enhanced spectrum sensing scheme under imperfect CSI conditions.

In response, we propose an IRS-assisted secure opportunistic access CRN, where an IRS is leveraged to enhance both spectrum sensing and secure transmission, simultaneously improving sensing accuracy and secrecy rates. We formulate a joint sensing and transmission security optimization problem to maximize the sum secrecy rate of SUs, accounting for both perfect and imperfect CSI scenarios. The proposed detection threshold ensures a specified probability of false alarm, and we employ a safe approximation to render the probability of detection tractable.

We utilize an efficient block coordinate descent (BCD)-based algorithm with successive convex approximation (SCA) and semidefinite relaxation (SDR) techniques to manage coupled variables. Particularly, the $\mathcal{S}$ -procedure transforms the infinite non-convex constraints into convex Linear Matrix Inequalities (LMIs) under the imperfect CSI scenario. Simulation results confirm that the proposed BCD-based algorithm rapidly converges, reduces optimal sensing time in spectrum sensing, and significantly enhances sensing accuracy and secrecy performance.

- To the best of our knowledge, this is the first work on studying an IRS-assisted secure opportunistic access CRNs where the IRS is exploited to assist both the spectrum sensing and the secure transmission. The joint sensing and secure transmission optimization problem is formulated under both the perfect and imperfect CSI cases for the maximization of the sum secrecy rate of SUs.

- Different from the benchmark schemes where the target probability of detection is achieved by the detection threshold, which can result in a high probability of false alarm and low spectrum usage, we exploit IRS as a novel approach to achieve the required probability of detection and set the detection threshold to ensure a low fixed probability of false alarm.

- For the perfect CSI case, the sum secrecy rate of SUs is maximized by jointly optimizing the sensing time, the beamforming of the SBS and the IRS phase shifts. The formulated problem is non-convex and challenging to tackle. An iterative BCD-based algorithm is proposed to optimize the beamforming and the phase shifts alternately for a given sensing time. Moreover, a safe approximation for Q -function is adopted to transform the probability of detection into a tractable expression. Furthermore, the SCA and SDR methods

are employed to address the non-convex terms and the rank-one condition, respectively.

- Considering a more practical scenario that the CSI of primary links and wiretap links cannot be perfectly known at the SBS, the secrecy rate maximization problem is studied under the worst-case bounded CSI error model. Especially, the design for IRS-enhanced spectrum sensing with bounded CSI is first studied. In this case, several equalities and the $\mathcal{S}$ -procedure are exploited to transform the infinite non-convex constraints into convex LMIs, and the BCD-based algorithm is adopted to optimize the beamforming design and the phase shifts alternately.

- Simulation results demonstrate that the proposed BCD-based algorithm converges within a few iterations in both cases. Moreover, by incorporating IRS into spectrum sensing, the optimal sensing time is significantly reduced, ensuring that both the required probability of detection and the probability of false alarm are achieved. This suggests that there is no need to consider a tradeoff between sensing and transmission. Furthermore, our proposed scheme substantially enhances sensing accuracy and secrecy rates in low SNR scenarios in both cases.

The rest of this paper is organized as follows. Section II introduces our proposed IRS-assisted opportunistic access CRNs. Section III presents the secrecy rate maximization under the perfect CSI case. Section IV presents the problem under the imperfect CSI case. Section V presents simulation results to evaluate the performance of the proposed schemes. Finally, the conclusion is presented in Section VI.

Notations: $\mathcal{CN}(\mu, \sigma^2)$ denotes a random variable that is circularly symmetric complex Gaussian distributed with μ mean and σ^2 variance. $\mathbb{C}^{A \times B}$ denotes the space of an $A \times B$ complex-valued matrix. $\mathbb{H}^N$ denotes the set of all N -dimensional complex Hermitian matrices. $\mathbf{I}_M$ denotes the $M \times M$ identity matrix. For a vector $\mathbf{a}$, and $\|\mathbf{a}\|$ denotes the Euclidean norm of $\mathbf{a}$. $\nabla_{\mathbf{a}} f(\mathbf{a})$ denotes the gradient vector of function $f(\mathbf{a})$ with respect to $\mathbf{a}$. For a matrix $\mathbf{A}$, $\text{Tr}(\mathbf{A})$, $\text{Rank}(\mathbf{A})$, $[\mathbf{A}]_{i,i}$, and $\|\mathbf{A}\|$ denote its trace, rank, (i,i) -th element, and Frobenius norm, respectively. $\mathbf{A}^T$ and $\mathbf{A}^H$ denote its transpose and conjugate transpose, respectively. $\mathbf{A} \succeq \mathbf{0}$ indicates that $\mathbf{A}$ is a positive semidefinite matrix. $\mathbf{A} \otimes \mathbf{B}$ is a Kronecker product of $\mathbf{A}$ and $\mathbf{B}$. $\text{vec}(\mathbf{A})$ is a column vector by stacking all the elements of $\mathbf{A}$. $Q(\cdot)$ is the Gaussian Q -function [46], i.e., $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty \exp(-\frac{t^2}{2}) dt$.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

As illustrated in Fig. 1, an IRS-assisted secure CR system is considered, which consists of a licensed primary network and an unlicensed secondary network. Specifically, the primary network consists of one single-antenna PU transmitter (PU-TX) and N single-antenna PU receivers (PU-RXs), while the secondary network comprises one secondary base station (SBS) with L antennas and K single-antenna SU receivers (SU-RXs). Considering a slotted time frame structure, in each frame duration T , SBS first performs spectrum sensing to detect the signals from the PU-TX in the sensing time slot τ

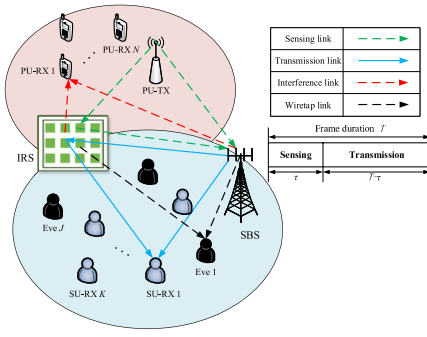


Fig. 1. The considered secure CR system model.

and then transmits data to K SU-RXs while restrains the interference caused to PU-RXs in the transmission time slot $T - \tau$. In the secondary network, J eavesdroppers (Eves) intercept the messages from the SBS. An IRS with M passive elements is exploited in both the sensing and transmission phases to enhance the sensing precision and the transmission security.

A quasi-static block fading channel model is considered. The channels from the PU-TX to the SBS and from the PU-TX to the IRS are denoted by $\mathbf{h}_d \in \mathbb{C}^{N \times 1}$ and $\mathbf{h}_r \in \mathbb{C}^{M \times 1}$, respectively. The uplink channel from the IRS to the SBS is denoted by $\mathbf{H} \in \mathbb{C}^{M \times N}$ and the downlink channel from the SBS to the IRS is denoted by $\mathbf{G} \in \mathbb{C}^{M \times N}$. Denoting the channels from the SBS to the k th SU-RX and from the IRS to the k th SU-RX by $\mathbf{g}_{d,k} \in \mathbb{C}^{N \times 1}$ and $\mathbf{g}_{r,k} \in \mathbb{C}^{M \times 1}$, respectively. The interference channels from the SBS to the n th PU-RX and from the IRS to the n th PU-RX are denoted by $\mathbf{I}_{d,n} \in \mathbb{C}^{N \times 1}$ and $\mathbf{I}_{r,n} \in \mathbb{C}^{M \times 1}$, respectively. The wiretap channels from the SBS to the j th Eve and from the IRS to the j th Eve are denoted by $\mathbf{f}_{d,j} \in \mathbb{C}^{N \times 1}$ and $\mathbf{f}_{r,j} \in \mathbb{C}^{M \times 1}$, respectively.

To realize opportunistic access, the SBS performs spectrum sensing and decides whether to access the primary band or not based on the received signal. The received signal at the SBS under two hypotheses can be expressed as

$$\mathcal{H}_1 : y_s(t) = \sqrt{P} (\mathbf{h}_d^H + \mathbf{h}_r^H \mathbf{\Psi} \mathbf{H}) x(t) + u_s(t), \quad (1a)$$

$$\mathcal{H}_0 : y_s(t) = u_s(t), \quad (1b)$$

where $\mathcal{H}_1$ denotes the presence of PU-TX, $\mathcal{H}_0$ denotes the absence of PU-TX, P is the transmit power of the PU-TX, $\mathbf{\Psi} = \text{diag}(\alpha_1 \exp(j\psi_1), \dots, \alpha_m \exp(j\psi_m), \dots, \alpha_M \exp(j\psi_M))$ denotes the diagonal matrix of the IRS reflecting coefficients, where α_m and ϕ_m are respectively the amplitude and the phase shifts of the m th reflecting element, $x(t)$ denotes the information symbol of the PU-TX, and $u_s(t) \sim \mathcal{CN}(0, \sigma_s^2)$ denotes the additive white Gaussian noise (AWGN) at the SBS. By adopting the energy detection scheme, the probability of detection P_d and the probability of false alarm P_f can be respectively expressed as [6]

$$P_d(\tau, \mathbf{\Psi}) = \Pr(y > \epsilon | \mathcal{H}_1) = Q \left(\frac{\epsilon - \sigma_s^2(L + \gamma)}{(\sigma_s^2 / \sqrt{\tau f_s}) \sqrt{2\gamma + L}} \right), \quad (2a)$$

$$P_f(\tau) = \Pr(y > \epsilon | \mathcal{H}_0) = Q \left(\frac{\epsilon - \sigma_s^2 L}{(\sigma_s^2 / \sqrt{\tau f_s}) \sqrt{L}} \right), \quad (2b)$$

where y is the decision statistic, ϵ is the detection threshold, $\gamma = \|\mathbf{h}_d^H + \mathbf{h}_r^H \mathbf{\Psi} \mathbf{H}\|^2 P / \sigma_s^2$ denotes the received SNR at the SBS and f_s denotes the sampling frequency.

Previous works required setting a detection threshold to achieve a target probability of detection. However, in low SNR regions, this approach may lead to a high probability of false alarm, causing inefficient spectrum usage [5], [6]. In this paper, a low fixed probability of false alarm $\bar{P}_f$ is considered to guarantee the utilization of spectrum resources, which can be achieved by setting the following detection threshold.

$$\tilde{\epsilon} = \frac{Q^{-1}(\bar{P}_f) \sigma_s^2 \sqrt{L}}{\sqrt{\tau f_s}} + \sigma_s^2 L, \quad (3)$$

where $Q^{-1}(\cdot)$ is the inverse of Gaussian Q -function [46]. As a remedy, we exploit the IRS as a novel approach to achieve the target probability of detection. Firstly, the probability of detection by using this approach is given by

$$\tilde{P}_d(\tau, \mathbf{\Psi}) = Q \left(\sqrt{\frac{L}{2\gamma + L}} Q^{-1}(\bar{P}_f) - \sqrt{\frac{\tau f_s}{2\gamma + L}} \gamma \right). \quad (4)$$

Then, how to design the IRS phase shifts to achieve the target probability of detection is given in the next section.

According to the principle of opportunistic access, the secondary transmission can be carried out in two cases. Case one is that the PU-TX is indeed inactive ($\mathcal{H}_0$) and the SBS determines to access the spectrum (no false alarm). In this case, the received signal at the k th SU-RX is given by

$$y_{r,k} = (\mathbf{g}_{d,k}^H + \mathbf{g}_{r,k}^H \mathbf{\Psi} \mathbf{G}) \sum_{k=1}^K \mathbf{w}_k s_k + u_{r,k}, \quad (5)$$

where $s_k \sim \mathcal{CN}(0, 1)$ denotes the data symbol for the k th SU-RX, $\mathbf{w}_k \in \mathbb{C}^{N \times 1}$ is the beamforming vector for the k th SU-RX and $u_{r,k} \sim \mathcal{CN}(0, \sigma_{r,k}^2)$ is the AWGN at k th SU-RX.

Case two is that the PU-TX is actually active ($\mathcal{H}_1$) but is not successfully detected by the SBS (miss detection). In this case, the signal received at k th SU-RX can be expressed as

$$y_{r,k} = (\mathbf{g}_{d,k}^H + \mathbf{g}_{r,k}^H \mathbf{\Psi} \mathbf{G}) \sum_{k=1}^K \mathbf{w}_k s_k + \sqrt{P} (h_{ps,k} + \mathbf{g}_{r,k}^H \mathbf{\Psi} \mathbf{h}_r) x + u_{r,k}, \quad (6)$$

where the second right-hand term is the interference signal from the PU-TX and $h_{ps,k} \in \mathbb{C}$ denotes the direct channel from the PU-TX to the k th SU-RX. In this case, the interference signal at the n th PU-RX can be expressed as

$$y_{p,n} = (\mathbf{I}_{d,n}^H + \mathbf{I}_{r,n}^H \mathbf{\Psi} \mathbf{G}) \sum_{k=1}^K \mathbf{w}_k s_k + u_{p,n}, \quad (7)$$

where $u_{p,n} \sim \mathcal{CN}(0, \sigma_{p,n}^2)$ is the AWGN at the n th PU-RX.

To facilitate the secure transmission, the worst-case assumption is considered [26]. In particular, all Eves are capable of eavesdropping the signals from the SBS and cancelling the multiuser interference. Thus, the signal received at j th Eve can be expressed as

$$y_{e,j} = (\mathbf{f}_{d,j}^H + \mathbf{f}_{r,j}^H \mathbf{\Psi} \mathbf{G}) \sum_{k=1}^K \mathbf{w}_k s_k + u_{e,j}, \quad (8)$$

where $u_{e,j} \sim \mathcal{CN}(0, \sigma_{e,j}^2)$ is the AWGN at the j th Eve.

The instantaneous achievable rate of k th SU-RX, denoted by $r_{0,k}$ for the case the PU-TX is inactive ($\mathcal{H}_0$) and by $r_{1,k}$ for the case the PU-TX is active ($\mathcal{H}_1$), are respectively given by

$$r_{0,k}(\tau, \Psi, \mathbf{w}_k) = \frac{T - \tau}{T} \Pr(\mathcal{H}_0)(1 - \bar{P}_f) \log_2(1 + \gamma_{0,k}), \quad (9a)$$

$$r_{1,k}(\tau, \Psi, \mathbf{w}_k) = \frac{T - \tau}{T} \Pr(\mathcal{H}_1)(1 - \tilde{P}_d) \log_2(1 + \gamma_{1,k}), \quad (9b)$$

where $\Pr(\mathcal{H}_0)$ and $\Pr(\mathcal{H}_1)$ are the inactive probability and active probability of PU-TX, respectively, $\gamma_{0,k}$ and $\gamma_{1,k}$ are the received signal-to-interference-and-noise ratio (SINR) at the k th SU-RX in both cases, which are shown in (10a) and (10b) at the bottom of this page, respectively.

Then, the average data rate of the j th Eve for decoding the message of the k th SU-RX can be expressed

$$r_{j,k}^E = \frac{T - \tau}{T} \left(\Pr(\mathcal{H}_0)(1 - \bar{P}_f) + \Pr(\mathcal{H}_1)(1 - \tilde{P}_d(\tau, \Psi)) \right) \times \log_2(1 + \gamma_{j,k}^E). \quad (11)$$

where $\gamma_{j,k}^E = \frac{|\mathbf{g}_{d,j}^H + \mathbf{g}_{r,j}^H \Psi \mathbf{G} \mathbf{w}_k|^2}{\sigma_{e,j}^2}$. Thus, the achievable secrecy rate of the k th SU-RX is given by

$$R_k^{\text{sec}}(\tau, \Psi, \mathbf{w}_k) = r_{0,k} + r_{1,k} - \max_{j \in \mathcal{J}} \{r_{j,k}^E\}. \quad (12)$$

To protect the quality of service (QoS) of the primary network, the minimum probability of detection constraint and the interference power constraint when the SBS misses the detection need to be considered, which are respectively expressed as

$$\tilde{P}_d(\tau, \Psi) \geq \bar{P}_d, \quad (13)$$

$$\frac{T - \tau}{T} \Pr(\mathcal{H}_1)(1 - \tilde{P}_d(\tau, \Psi)) \sum_{k=1}^K \left| (\mathbf{I}_{d,n}^H + \mathbf{I}_{r,n}^H \Psi \mathbf{G}) \mathbf{w}_k \right|^2 \leq \Gamma_n, \forall n, \quad (14)$$

where $\bar{P}_d$ is the target probability of detection and Γ_n is the maximum tolerable interference at the n th PU-RX.

To protect the SBS, the total transmit power constraint for the SBS is given by

$$\frac{T - \tau}{T} \left(\Pr(\mathcal{H}_0)(1 - \bar{P}_f) + \Pr(\mathcal{H}_1)(1 - \tilde{P}_d(\tau, \Psi)) \right) \times \sum_{k=1}^K \|\mathbf{w}_k\|^2 \leq P_T, \quad (15)$$

where P_T is the maximum transmit power budget of the SBS.

B. Problem Formulation

In contrast to the previous works where the IRS only assists the secure transmission [42], [43], [44], [45], in this paper, the IRS is exploited to assist both the spectrum sensing and the secure transmission. The objective is to maximize the sum secrecy rate of all the SU-RXs, i.e., $\sum_{k=1}^K R_k^{\text{sec}}$, by optimizing the sensing time, the beamforming of the SBS and the phase shifts of IRS, which is given by

$$\mathbf{P}_1 : \max_{\mathbf{w}_k, \Psi, \tau} \sum_{k=1}^K R_k^{\text{sec}}(\tau, \Psi, \mathbf{w}_k) \quad (16a)$$

$$\text{s.t. } C1 : 0 \leq \tau \leq T, \quad (16b)$$

$$C2 : 0 \leq |[\Psi]_{m,m}| \leq 1, \forall m, \quad (16c)$$

$$(13) - (15), \quad (16d)$$

where $C1$ is the sensing time constraint, $C2$ is the constraint of the IRS passive gain, the constraint (13) guarantees the target probability of detection, the constraint (14) limits the maximum interference leakage to PU-RXs when the SBS misses the detection, the constraint (15) is served as the total power constraint for the SBS. Problem $\mathbf{P}_1$ is a highly non-convex problem due to the coupling among the variables τ , $\mathbf{w}_k$ and Ψ . To overcome this challenge, an exhaustive search [5] is adopted to find the optimal sensing time τ in the outer layer. Then, in the inner layer, a BCD-based algorithm is proposed to optimize the beamforming of the SBS and the IRS phase shifts alternately based on a given sensing time.

III. SECRECY RATE MAXIMIZATION UNDER PERFECT CSI

In this section, the proposed IRS-assisted secure CRNs is studied under the perfect CSI. The assumption of the full CSI of eavesdropping channels accessed at the SBS is ideal. However, this is realizable since Eves can be treated as the other users in the secondary network but are not trusted by the SU-RXs, and the CSI of eavesdropping links can be obtained at the SBS [42].

A. Beamforming Optimization for Given IRS Phase Shifts

For any given Ψ , the optimization problem for the beamforming of the SBS is given by

$$\mathbf{P}_2 : \max_{\mathbf{w}_k} \sum_{k=1}^K R_k^{\text{sec}}(\mathbf{w}_k) \quad (17a)$$

$$\text{s.t. } (14), (15). \quad (17b)$$

By defining $\mathbf{W}_k = \mathbf{w}_k \mathbf{w}_k^H \in \mathbb{H}^L$, we respectively rewrite the quadratic term $|\mathbf{g}_{d,k}^H + \mathbf{g}_{r,k}^H \Psi \mathbf{G} \mathbf{w}_k|^2$,

$$\gamma_{0,k} = \frac{|\mathbf{g}_{d,k}^H + \mathbf{g}_{r,k}^H \Psi \mathbf{G} \mathbf{w}_k|^2}{\sum_{j \neq k}^K |\mathbf{g}_{d,k}^H + \mathbf{g}_{r,k}^H \Psi \mathbf{G} \mathbf{w}_j|^2 + \sigma_{r,k}^2}, \quad (10a)$$

$$\gamma_{1,k} = \frac{|\mathbf{g}_{d,k}^H + \mathbf{g}_{r,k}^H \Psi \mathbf{G} \mathbf{w}_k|^2}{\sum_{j \neq k}^K |\mathbf{g}_{d,k}^H + \mathbf{g}_{r,k}^H \Psi \mathbf{G} \mathbf{w}_j|^2 + P|h_{ps,k} + \mathbf{g}_{r,k}^H \Psi \mathbf{h}_r|^2 + \sigma_{r,k}^2}. \quad (10b)$$

$|h_{ps,k} + \mathbf{g}_{r,k}^H \Psi \mathbf{h}_r|^2$ and $|\mathbf{f}_{d,j}^H + \mathbf{f}_{d,j}^H \Psi \mathbf{G}|^2$ in the objective function and $|(\mathbf{I}_{d,n}^H + \mathbf{I}_{d,n}^H \Psi \mathbf{G}) \mathbf{w}_k|^2$ in (14) as

$$|(\mathbf{g}_{d,k}^H + \mathbf{g}_{r,k}^H \Psi \mathbf{G}) \mathbf{w}_k|^2 = |\mathbf{e}^H \mathbf{G}_k \mathbf{w}_k|^2 = \text{Tr}(\mathbf{W}_k \mathbf{G}_k^H \mathbf{e} \mathbf{e}^H \mathbf{G}_k), \quad (18a)$$

$$|h_{ps,k} + \mathbf{g}_{r,k}^H \Psi \mathbf{h}_r|^2 = |h_{ps,k} + \mathbf{e}^H \mathbf{q}_k|^2 = |z_k|^2, \quad (18b)$$

$$|\mathbf{f}_{d,j}^H + \mathbf{f}_{d,j}^H \Psi \mathbf{G}|^2 = |\mathbf{e}^H \mathbf{F}_j \mathbf{w}_k|^2 = \text{Tr}(\mathbf{W}_k \mathbf{F}_j^H \mathbf{e} \mathbf{e}^H \mathbf{F}_j), \quad (18c)$$

$$|(\mathbf{I}_{d,n}^H + \mathbf{I}_{r,n}^H \Psi \mathbf{G}) \mathbf{w}_k|^2 = |\mathbf{e}^H \mathbf{V}_n \mathbf{w}_k|^2 = \text{Tr}(\mathbf{W}_k \mathbf{V}_n^H \mathbf{e} \mathbf{e}^H \mathbf{V}_n), \quad (18d)$$

where $\mathbf{e} = [\alpha_1 \exp(j\psi_1), \dots, \alpha_M \exp(j\psi_M) \ 1]^T$, $\mathbf{G}_k = \begin{bmatrix} \text{diag}(\mathbf{g}_{r,k}^H) \mathbf{G} \\ \mathbf{g}_{d,k}^H \end{bmatrix}$, $\mathbf{q}_k = \text{diag}(\mathbf{g}_{r,k}^H) \mathbf{h}_r$, $z_k = h_{ps,k} + \mathbf{e}^H \mathbf{q}_k$, $\mathbf{F}_j = \begin{bmatrix} \text{diag}(\mathbf{f}_{r,j}^H) \mathbf{G} \\ \mathbf{f}_{d,j}^H \end{bmatrix}$ and $\mathbf{V}_n = \begin{bmatrix} \text{diag}(\mathbf{I}_{r,n}^H) \mathbf{G} \\ \mathbf{I}_{d,n}^H \end{bmatrix}$. Then, $\gamma_{0,k}$, $\gamma_{1,k}$ and $\gamma_{j,k}^E$ can be respectively rewritten as

$$\gamma_{0,k} = \frac{\text{Tr}(\mathbf{W}_k \mathbf{G}_k^H \mathbf{e} \mathbf{e}^H \mathbf{G}_k)}{\sum_{j \neq k}^K \text{Tr}(\mathbf{W}_j \mathbf{G}_k^H \mathbf{e} \mathbf{e}^H \mathbf{G}_k) + \sigma_{r,k}^2}, \quad (19a)$$

$$\gamma_{1,k} = \frac{\text{Tr}(\mathbf{W}_k \mathbf{G}_k^H \mathbf{e} \mathbf{e}^H \mathbf{G}_k)}{\sum_{j \neq k}^K \text{Tr}(\mathbf{W}_j \mathbf{G}_k^H \mathbf{e} \mathbf{e}^H \mathbf{G}_k) + P|z_k|^2 + \sigma_{r,k}^2}, \quad (19b)$$

$$\gamma_{j,k}^E = \frac{\text{Tr}(\mathbf{W}_k \mathbf{F}_j^H \mathbf{e} \mathbf{e}^H \mathbf{F}_j)}{\sigma_{e,j}^2}. \quad (19c)$$

Thus, the problem $\mathbf{P}_2$ is reformulated as

$$\mathbf{P}_{2.1} : \max_{\mathbf{W}_k \in \mathbb{H}^N} a_0 \sum_{k=1}^K \log_2(1 + \gamma_{0,k}) + a_1 \sum_{k=1}^K \log_2(1 + \gamma_{1,k}) - (a_0 + a_1) \sum_{k=1}^K \max_{j \in \mathcal{J}} \{\log_2(1 + \gamma_{j,k}^E)\} \quad (20a)$$

$$\text{s.t. } (a_0 + a_1) \sum_{k=1}^K \text{Tr}(\mathbf{W}_k) \leq P_T, \quad (20b)$$

$$a_1 \sum_{k=1}^K \text{Tr}(\mathbf{W}_k \mathbf{V}_n^H \mathbf{e} \mathbf{e}^H \mathbf{V}_n) \leq \Gamma_n, \forall n, \quad (20c)$$

$$\text{Rank}(\mathbf{W}_k) \leq 1, \forall k, \quad (20d)$$

$$\mathbf{W}_k \succeq \mathbf{0}, \forall k, \quad (20e)$$

where $a_0 = (1 - \tau/T) \Pr(\mathcal{H}_0)(1 - \bar{P}_f)$, $a_1(\Psi) = (1 - \tau/T) \Pr(\mathcal{H}_1)(1 - \bar{P}_d(\Psi))$, and the constraints (20d) and (20e) are imposed to ensure that $\mathbf{W}_k = \mathbf{w}_k \mathbf{w}_k^H$ holds. The problem $\mathbf{P}_{2.1}$ is non-convex due to the objective function and the rank-one constraint in (20d). By introducing the auxiliary variables $\delta_0 = [\delta_{0,1}, \dots, \delta_{0,K}]$, $\delta_1 = [\delta_{1,1}, \dots, \delta_{1,K}]$ and $\kappa = [\kappa_1, \dots, \kappa_K]$, we rewrite the

$\mathbf{P}_{2.2}$:

$$\max_{\mathbf{W}_k, \delta_0, \delta_1, \kappa} a_0 \sum_{k=1}^K \log_2(1 + \delta_{0,k}) + a_1 \sum_{k=1}^K \log_2(1 + \delta_{1,k}) - (a_0 + a_1) \sum_{k=1}^K \log_2(1 + \kappa_k) \quad (21a)$$

$$\text{s.t. } (20b) - (20e), \quad (21b)$$

$$\gamma_{0,k} \geq \delta_{0,k}, \forall k, \quad (21c)$$

$$\gamma_{1,k} \geq \delta_{1,k}, \forall k, \quad (21d)$$

$$\max_{j \in \mathcal{J}} \{\gamma_{j,k}^E\} \leq \kappa_k, \forall k. \quad (21e)$$

The non-convexity in the problem $\mathbf{P}_{2.2}$ originates from the objective function and the constraints (20d), (21c) and (21d). For a given feasible point κ_k^ℓ at the ℓ th iteration of the SCA, the lower bound of the term $f(\kappa_k) \triangleq \sum_{k=1}^K \log_2(1 + \kappa_k)$ in the objective function can be obtained by exploiting the first-order Taylor approximation, which is given by

$$f(\kappa_k) \geq f(\kappa_k^\ell) + \sum_{k=1}^K \nabla_{\kappa_k} f(\kappa_k^\ell) (\kappa_k - \kappa_k^\ell) \triangleq \hat{f}(\kappa_k). \quad (22)$$

Then, by introducing a set of auxiliary variables $\zeta = [\zeta_1, \dots, \zeta_k, \dots, \zeta_K]$, the constraint (21c) can be transformed into

$$\text{Tr}(\mathbf{W}_k \mathbf{G}_k^H \mathbf{e} \mathbf{e}^H \mathbf{G}_k) \geq \delta_{0,k} \zeta_k, \quad (23a)$$

$$\sum_{j \neq k}^K \text{Tr}(\mathbf{W}_j \mathbf{G}_k^H \mathbf{e} \mathbf{e}^H \mathbf{G}_k) + \sigma_{r,k}^2 \leq \zeta_k. \quad (23b)$$

Note that the right hand term of (23a) is the joint convex function with respect to $\delta_{0,k}$ and ζ_k , utilizing the SCA, $\delta_{0,k} \zeta_k$ can be rewritten as

$$\delta_{0,k} \zeta_k \geq \delta_{0,k}^\ell \zeta_k^\ell + \delta_{0,k}^\ell (\zeta_k - \zeta_k^\ell) + \zeta_k^\ell (\delta_{0,k} - \delta_{0,k}^\ell) \triangleq \hat{f}(\delta_{0,k}, \zeta_k), \quad (24)$$

where $(\delta_{0,k}^\ell, \zeta_k^\ell)$ is a given feasible point at the ℓ th iteration of the SCA.

Similarly, by introducing the auxiliary variables $\xi = [\xi_1, \dots, \xi_k, \dots, \xi_K]$, the constraint (21d) is transformed into

$$\text{Tr}(\mathbf{W}_k \mathbf{G}_k^H \mathbf{e} \mathbf{e}^H \mathbf{G}_k) \geq \delta_{1,k} \xi_k, \quad (25a)$$

$$\sum_{j \neq k}^K \text{Tr}(\mathbf{W}_j \mathbf{G}_k^H \mathbf{e} \mathbf{e}^H \mathbf{G}_k) + P|z_k|^2 + \sigma_{r,k}^2 \leq \xi_k. \quad (25b)$$

Then, the lower bound of $\delta_{1,k} \xi_k$ in (25a) is given by

$$\delta_{1,k} \xi_k \geq \delta_{1,k}^\ell \xi_k^\ell + \delta_{1,k}^\ell (\xi_k - \xi_k^\ell) + \xi_k^\ell (\delta_{1,k} - \delta_{1,k}^\ell) \triangleq \hat{f}(\delta_{1,k}, \xi_k), \quad (26)$$

where $(\delta_{1,k}^\ell, \xi_k^\ell)$ is a given feasible point at the ℓ th iteration of the SCA. As a result, the problem $\mathbf{P}_{2.2}$ is

reformulated as

$$\mathbf{P}_{2.3} : \max_{\mathbf{W}_k, \delta_0, \delta_1, \kappa, \zeta, \xi} a_0 \sum_{k=1}^K \log_2(1 + \delta_{0,k}) + a_1 \sum_{k=1}^K \log_2(1 + \delta_{1,k}) - (a_0 + a_1) \hat{f}(\kappa_k) \quad (27a)$$

$$\text{s.t. (20b) - (20e),} \quad (27b)$$

$$\text{Tr}(\mathbf{W}_k \mathbf{G}_k^H \mathbf{e} \mathbf{e}^H \mathbf{G}_k) \geq \hat{f}(\delta_{0,k}, \zeta_k), \forall k, \quad (27c)$$

$$\sum_{j \neq k}^K \text{Tr}(\mathbf{W}_j \mathbf{G}_k^H \mathbf{e} \mathbf{e}^H \mathbf{G}_k) + \sigma_{r,k}^2 \leq \zeta_k, \forall k, \quad (27d)$$

$$\text{Tr}(\mathbf{W}_k \mathbf{G}_k^H \mathbf{e} \mathbf{e}^H \mathbf{G}_k) \geq \hat{f}(\delta_{1,k}, \xi_k), \forall k, \quad (27e)$$

$$\sum_{j \neq k}^K \text{Tr}(\mathbf{W}_j \mathbf{G}_k^H \mathbf{e} \mathbf{e}^H \mathbf{G}_k) + P|z_k|^2 + \sigma_{r,k}^2 \leq \xi_k, \forall k, \quad (27f)$$

$$\max_{j \in \mathcal{J}} \{\gamma_{j,k}^E\} \leq \kappa_k, \forall k. \quad (27g)$$

As for the rank-one constraint (20d), it can be relaxed by employing the SDR method. Finally, the relaxed version of the problem $\mathbf{P}_{2.3}$ is a standard convex semidefinite program (SDP) problem, which can be solved via CVX [47]. The similar proof for the tightness of the SDR in $\mathbf{P}_{2.3}$ can be found in [26]. Besides, the obtained value of the objective function in $\mathbf{P}_{2.3}$ is served as a lower bound of the original problem $\mathbf{P}_2$ due to exploitation of the SCA. Then, the optimal beamforming of the SBS can be obtained by decomposing the solution of $\mathbf{P}_{2.3}$ with the eigenvalue decomposition.

B. IRS Phase Shifts Optimization for Given Beamforming

For any given $\mathbf{w}_k$, the optimization problem for the IRS phase shifts is given by

$$\mathbf{P}_3 : \max_{\Psi} \sum_{k=1}^K R_k^{\text{sec}}(\Psi) \quad (28a)$$

$$\text{s.t. } C2, (13) - (15). \quad (28b)$$

The problem $\mathbf{P}_3$ is non-convex with respect to Ψ . Let $\mathbf{E} = \mathbf{e} \mathbf{e}^H \in \mathbb{C}^{M+1}$. The term $\left\| \mathbf{h}_d^H + \mathbf{h}_r^H \Psi \mathbf{H} \right\|^2$ in the probability of detection $\tilde{P}_d$ can be rewritten as

$$\left\| \mathbf{h}_d^H + \mathbf{h}_r^H \Psi \mathbf{H} \right\|^2 = \left\| \mathbf{e}^H \mathbf{H}_s \right\|^2 = \text{Tr}(\mathbf{E} \mathbf{H}_s \mathbf{H}_s^H), \quad (29)$$

where $\mathbf{e} = [\alpha_1 \exp(j\psi_1), \dots, \alpha_M \exp(j\psi_M), 1]^T$ and $\mathbf{H}_s = \begin{bmatrix} \text{diag}(\mathbf{h}_r^H) \mathbf{H} \\ \mathbf{h}_d^H \end{bmatrix}$. Thus, the probability of detection can be rewritten as

$$\tilde{P}_d(\mathbf{E}) = Q \left(\frac{\tilde{\epsilon} - \sigma_s^2(L + \text{Tr}(\mathbf{E} \mathbf{H}_s \mathbf{H}_s^H) P / \sigma_s^2)}{(\sigma_s^2 / \sqrt{\tau f_s}) \sqrt{2 \text{Tr}(\mathbf{E} \mathbf{H}_s \mathbf{H}_s^H) P / \sigma_s^2 + L}} \right). \quad (30)$$

Then, we rewrite $\gamma_{0,k}$, $\gamma_{1,k}$ and $\gamma_{j,k}^E$ as

$$\gamma_{0,k} = \frac{\text{Tr}(\mathbf{E} \mathbf{G}_k \mathbf{W}_k \mathbf{G}_k^H)}{\sum_{j \neq k}^K \text{Tr}(\mathbf{E} \mathbf{G}_k \mathbf{W}_j \mathbf{G}_k^H) + \sigma_{r,k}^2}, \quad (31a)$$

$$\gamma_{1,k} = \frac{\text{Tr}(\mathbf{E} \mathbf{G}_k \mathbf{W}_k \mathbf{G}_k^H)}{\sum_{j \neq k}^K \text{Tr}(\mathbf{E} \mathbf{G}_k \mathbf{W}_j \mathbf{G}_k^H) + P \text{Tr}(\mathbf{E} \mathbf{R}_k \mathbf{R}_k^H) + \sigma_{r,k}^2}, \quad (31b)$$

$$\gamma_{j,k}^E = \frac{\text{Tr}(\mathbf{E} \mathbf{F}_j \mathbf{W}_k \mathbf{F}_j^H)}{\sigma_{e,j}^2}. \quad (31c)$$

Recall that $\mathbf{W}_k = \mathbf{w}_k \mathbf{w}_k^H$, $\mathbf{G}_k = \begin{bmatrix} \text{diag}(\mathbf{g}_{r,k}^H) \mathbf{G} \\ \mathbf{g}_{d,k}^H \end{bmatrix}$, $\mathbf{q}_k = \text{diag}(\mathbf{g}_{r,k}^H) \mathbf{h}_r$, $z_k = h_{\text{ps},k} + \mathbf{e}^H \mathbf{q}_k$, $\mathbf{F}_j = \begin{bmatrix} \text{diag}(\mathbf{f}_{r,j}^H) \mathbf{G} \\ \mathbf{f}_{d,j}^H \end{bmatrix}$, $\mathbf{V}_n = \begin{bmatrix} \text{diag}(\mathbf{I}_{r,n}^H) \mathbf{G} \\ \mathbf{I}_{d,n}^H \end{bmatrix}$, $a_0 = (1 - \tau/T) \Pr(\mathcal{H}_0)(1 - \bar{P}_f)$, $a_1 = (1 - \tau/T) \Pr(\mathcal{H}_1)(1 - \tilde{P}_d(\mathbf{E}))$, and define $\mathbf{R}_k = \begin{bmatrix} \mathbf{q}_k \\ h_{\text{ps},k} \end{bmatrix}$, the problem $\mathbf{P}_3$ can be rewritten as

$$\mathbf{P}_{3.1} : \max_{\mathbf{E} \in \mathbb{C}^{M+1}} a_0 \sum_{k=1}^K \log_2(1 + \gamma_{0,k}) + a_1 \sum_{k=1}^K \log_2(1 + \gamma_{1,k}) - (a_0 + a_1) \sum_{k=1}^K \max_{j \in \mathcal{J}} \{\log_2(1 + \gamma_{j,k}^E)\} \quad (32a)$$

$$\text{s.t. } \tilde{P}_d(\mathbf{E}) \geq \bar{P}_d, \quad (32b)$$

$$(a_0 + a_1) \sum_{k=1}^K \text{Tr}(\mathbf{W}_k) \leq P_T, \quad (32c)$$

$$a_1 \sum_{k=1}^K \text{Tr}(\mathbf{E} \mathbf{V}_n \mathbf{W}_k \mathbf{V}_n^H) \leq \Gamma_n, \forall n, \quad (32d)$$

$$\text{Rank}(\mathbf{E}) = 1, \quad (32e)$$

$$\mathbf{E} \succeq \mathbf{0}, \quad (32f)$$

$$[\mathbf{E}]_{m,m} \leq 1, \forall m, \quad (32g)$$

where the constraints (32e) and (32f) are imposed to ensure that $\mathbf{E} = \mathbf{e} \mathbf{e}^H$ holds. Note that in the objective function and the constraint (32d), $\mathbf{E}$ exists in both the Q -function of $\tilde{P}_d$ and the channel matrix trace, which is challenging to handle. Thus, we introduce a safe approximation for the Q -function, which is $Q(x) \leq 1/12 \exp(-x^2/2) + 1/4 \exp(-2x^2/3)$. This approximation is very tight for $x > \sqrt{2}/2$ [48]. Typically, the probability of detection must satisfy $\tilde{P}_d \geq 0.9$. Let $q(\mathbf{E}) \triangleq \frac{\tilde{\epsilon} - \sigma_s^2(L + \text{Tr}(\mathbf{E} \mathbf{H}_s \mathbf{H}_s^H) P / \sigma_s^2)}{(\sigma_s^2 / \sqrt{\tau f_s}) \sqrt{2 \text{Tr}(\mathbf{E} \mathbf{H}_s \mathbf{H}_s^H) P / \sigma_s^2 + L}}$, when $\tilde{P}_d(q) \geq 0.9$, q is negative and $q < -\sqrt{2}/2$. Thus, the equality $1 - Q(q) = Q(-q)$ is also adopted to ensure the tightness condition $q > \sqrt{2}/2$ holds and the target probability of detection can be achieved. Then, we transform the term $1 - \tilde{P}_d(\mathbf{E})$ into

$$1 - P_d(\mathbf{E}) \leq \frac{1}{12} \exp\left(-\frac{1}{2} \hat{y}\right) + \frac{1}{4} \exp\left(-\frac{2}{3} \hat{y}\right) \triangleq g(\hat{y}), \quad (33)$$

where $\hat{y}$ is the auxiliary variable, which is constrained by

$$\sqrt{\hat{y}} \leq \frac{\sigma_s^2(L + \text{Tr}(\mathbf{E}\mathbf{H}_s\mathbf{H}_s^H)P/\sigma_s^2) - \epsilon}{(\sigma_s^2/\sqrt{\tau f_s})\sqrt{2\text{Tr}(\mathbf{E}\mathbf{H}_s\mathbf{H}_s^H)P/\sigma_s^2 + L}}. \quad (34)$$

Then, we further introduce the auxiliary variables $\hat{\delta}_0 = [\hat{\delta}_{0,1}, \dots, \hat{\delta}_{0,K}]$, $\hat{\delta}_1 = [\hat{\delta}_{1,1}, \dots, \hat{\delta}_{1,K}]$ and $\hat{\kappa} = [\hat{\kappa}_1, \dots, \hat{\kappa}_K]$. The problem $\mathbf{P}_{3.1}$ is reformulated as

$\mathbf{P}_{3.2}$:

$$\max_{\mathbf{E}, \hat{y}, \hat{\delta}_0, \hat{\delta}_1, \hat{\kappa}} a_0 \sum_{k=1}^K \log_2(1 + \hat{\delta}_{0,k}) + \alpha g(\hat{y}) \sum_{k=1}^K \log_2(1 + \hat{\delta}_{1,k}) - (a_0 + \alpha g(\hat{y})) \sum_{k=1}^K \log_2(1 + \hat{\kappa}_k) \quad (35a)$$

$$\text{s.t. (32e) - (32g), (34)} \quad (35b)$$

$$g(\hat{y}) \leq 1 - \bar{P}_d, \quad (35c)$$

$$(a_0 + \alpha g(\hat{y})) \sum_{k=1}^K \text{Tr}(\mathbf{W}_k) \leq P_T, \quad (35d)$$

$$\alpha g(\hat{y}) \sum_{k=1}^K \text{Tr}(\mathbf{E}\mathbf{V}_n\mathbf{W}_k\mathbf{V}_n^H) \leq \Gamma_n, \forall n, \quad (35e)$$

$$\max_{j \in \mathcal{J}} \left\{ \frac{\text{Tr}(\mathbf{E}\mathbf{F}_j\mathbf{W}_k\mathbf{F}_j^H)}{\sigma_{e,j}^2} \right\} \leq \hat{\kappa}_k, \forall k, \quad (35f)$$

$$\gamma_{0,k} \geq \hat{\delta}_{0,k}, \forall k, \quad (35g)$$

$$\gamma_{1,k} \geq \hat{\delta}_{1,k}, \forall k, \quad (35h)$$

where $\alpha = (1 - \tau/T) \Pr(\mathcal{H}_1)$. The non-convexity in the problem $\mathbf{P}_{3.2}$ originates from the objective function and the constraints (32e), (34), (35e), (35g) and (35h). Let $f(\hat{y}, \hat{\delta}_{1,k}) \triangleq g(\hat{y}) \sum_{k=1}^K \log_2(1 + \hat{\delta}_{1,k})$, the lower bound of $f(\hat{y}, \hat{\delta}_{1,k})$ for a given feasible point $(\hat{y}^\ell, \hat{\delta}_{1,k}^\ell)$ at the ℓ th iteration of the SCA is given by

$$f(\hat{y}, \hat{\delta}_{1,k}) \geq f(\hat{y}^\ell, \hat{\delta}_{1,k}^\ell) + \sum_{k=1}^K \nabla_{\hat{\delta}_{1,k}} f(\hat{y}^\ell, \hat{\delta}_{1,k}^\ell)(\hat{\delta}_{1,k} - \hat{\delta}_{1,k}^\ell) + \nabla_{\hat{y}} f(\hat{y}^\ell, \hat{\delta}_{1,k}^\ell)(\hat{y} - \hat{y}^\ell) \triangleq \hat{f}(\hat{y}, \hat{\delta}_{1,k}). \quad (36)$$

Similarly, the lower bound of $f(\hat{y}, \hat{\kappa}_k) \triangleq g(\hat{y}) \sum_{k=1}^K \log_2(1 + \hat{\kappa}_k)$ for a given feasible point $(\hat{y}^\ell, \hat{\kappa}_k^\ell)$ at the ℓ th iteration of the SCA is given by

$$f(\hat{y}, \hat{\kappa}_k) \geq f(\hat{y}^\ell, \hat{\kappa}_k^\ell) + \sum_{k=1}^K \nabla_{\hat{\kappa}_k} f(\hat{y}^\ell, \hat{\kappa}_k^\ell)(\hat{\kappa}_k - \hat{\kappa}_k^\ell) + \nabla_{\hat{y}} f(\hat{y}^\ell, \hat{\kappa}_k^\ell)(\hat{y} - \hat{y}^\ell) \triangleq \hat{f}(\hat{y}, \hat{\kappa}_k). \quad (37)$$

For the constraint (35e), the left hand term $f(\hat{y}, \mathbf{E}) \triangleq g(\hat{y}) \sum_{k=1}^K \text{Tr}(\mathbf{E}\mathbf{V}_n\mathbf{W}_k\mathbf{V}_n^H)$ is the joint convex function with respect to $\hat{y}$ and $\mathbf{E}$, its lower bound at the ℓ th iteration of the SCA for a given feasible point $(\hat{y}^\ell, \mathbf{E}^\ell)$ is

given by

$$f(\hat{y}, \mathbf{E}) \geq f(\hat{y}^\ell, \mathbf{E}^\ell) + \sum_{k=1}^K \text{Tr}(\nabla_{\mathbf{E}} f(\hat{y}^\ell, \mathbf{E}^\ell)^H (\mathbf{E} - \mathbf{E}^\ell)) + \nabla_{\hat{y}} f(\hat{y}^\ell, \mathbf{E}^\ell)(\hat{y} - \hat{y}^\ell) \triangleq \hat{f}(\hat{y}, \mathbf{E}). \quad (38)$$

To deal with constraint (34), we equivalently rewrite it as

$$\sqrt{\hat{y}} \left(\frac{\sigma_s^2}{\sqrt{\tau f_s}} \sqrt{2\text{Tr}(\mathbf{E}\mathbf{H}_s\mathbf{H}_s^H)P/\sigma_s^2 + L} - \sigma_s^2(L + \text{Tr}(\mathbf{E}\mathbf{H}_s\mathbf{H}_s^H)P/\sigma_s^2) + \epsilon \leq 0. \quad (39)$$

Then, for a given feasible point $(\hat{y}^\ell, \mathbf{E}^\ell)$, the upper bound of $u(\hat{y}, \mathbf{E}) \triangleq \sqrt{\hat{y}} \sqrt{2\text{Tr}(\mathbf{E}\mathbf{H}_s\mathbf{H}_s^H)P/\sigma_s^2 + L}$ at the ℓ th iteration of the SCA can be given by

$$u(\hat{y}, \mathbf{E}) \leq u(\hat{y}^\ell, \mathbf{E}^\ell) + \sum_{k=1}^K \text{Tr}(\nabla_{\mathbf{E}} u(\hat{y}^\ell, \mathbf{E}^\ell)^H (\mathbf{E} - \mathbf{E}^\ell)) + \nabla_{\hat{y}} u(\hat{y}^\ell, \mathbf{E}^\ell)(\hat{y} - \hat{y}^\ell) \triangleq \hat{u}(\hat{y}, \mathbf{E}), \quad (40)$$

where $\nabla_{\mathbf{E}} u(\hat{y}^\ell, \mathbf{E}^\ell)$ and $\nabla_{\hat{y}} u(\hat{y}^\ell, \mathbf{E}^\ell)$ can be respectively expressed as

$$\nabla_{\mathbf{E}} u(\hat{y}^\ell, \mathbf{E}^\ell) = \frac{\sqrt{\hat{y}^\ell} \mathbf{P}\mathbf{H}_s\mathbf{H}_s^H}{\sqrt{2\text{Tr}(\mathbf{E}^\ell\mathbf{H}_s\mathbf{H}_s^H)P/\sigma_s^2 + N\sigma_s^4}}, \quad (41a)$$

$$\nabla_{\hat{y}} u(\hat{y}^\ell, \mathbf{E}^\ell) = \frac{\sqrt{2\text{Tr}(\mathbf{E}^\ell\mathbf{H}_s\mathbf{H}_s^H)P/\sigma_s^2 + L}}{2\sqrt{\hat{y}^\ell}}. \quad (41b)$$

As for the remaining constraints, similar to the methods adopted for solving the problem $\mathbf{P}_{2.2}$, we adopt the SDR method to relax the constraint (32e) and introducing auxiliary variables $\hat{\zeta}_k$ and $\hat{\xi}_k$ to resolve (35g) and (35h), respectively. Then, the problem $\mathbf{P}_{3.2}$ can be transformed into

$$\mathbf{P}_{3.3} : \max_{\mathbf{E}, \hat{y}, \hat{\delta}_0, \hat{\delta}_1, \hat{\kappa}} a_0 \sum_{k=1}^K \log_2(1 + \hat{\delta}_{0,k}) + \alpha \hat{f}(\hat{y}, \hat{\delta}_{1,k}) - a_0 \hat{f}(\hat{\kappa}_k) - \alpha \hat{f}(\hat{y}, \hat{\kappa}_k) \quad (42a)$$

$$\text{s.t. (32f), (32g), (35c), (35d), (35f),} \quad (42b)$$

$$\alpha \hat{f}(\hat{y}, \mathbf{E}) \leq \Gamma_n, \forall n, \quad (42c)$$

$$\hat{u}(\hat{y}, \mathbf{E}) - \sigma_s^2(L + \text{Tr}(\mathbf{E}\mathbf{H}_s\mathbf{H}_s^H)P/\sigma_s^2) + \epsilon \leq 0, \quad (42d)$$

$$\text{Tr}(\mathbf{E}\mathbf{G}_k\mathbf{W}_k\mathbf{G}_k^H) \geq \hat{\delta}_{0,k}\hat{\zeta}_k, \forall k, \quad (42e)$$

$$\sum_{j \neq k}^K \text{Tr}(\mathbf{E}\mathbf{G}_k\mathbf{W}_j\mathbf{G}_k^H) + \sigma_{r,k}^2 \leq \hat{\zeta}_k, \forall k, \quad (42f)$$

$$\text{Tr}(\mathbf{E}\mathbf{G}_k\mathbf{W}_k\mathbf{G}_k^H) \geq \hat{\delta}_{1,k}\hat{\xi}_k, \forall k, \quad (42g)$$

$$\sum_{j \neq k}^K \text{Tr}(\mathbf{E}\mathbf{G}_k\mathbf{W}_j\mathbf{G}_k^H) + P\text{Tr}(\mathbf{E}\mathbf{R}_k\mathbf{R}_k^H) + \sigma_{r,k}^2 \leq \hat{\xi}_k, \forall k, \quad (42h)$$

where $f(\hat{\kappa}_k)$ can be given by (22) by replacing κ_k with $\hat{\kappa}_k$. Note that the right-hand terms of (42e) and (42g) are still coupled, the lower bounds of $\hat{\delta}_{0,k}\hat{\zeta}_k$ and $\hat{\delta}_{1,k}\hat{\xi}_k$ for a given feasible point at the ℓ th iteration of the SCA are respectively given by

$$\begin{aligned} \hat{\delta}_{0,k}\hat{\zeta}_k &\geq \hat{\delta}_{0,k}^{\ell}\hat{\zeta}_k^{\ell} + \hat{\delta}_{0,k}^{\ell}(\hat{\zeta}_k - \hat{\zeta}_k^{\ell}) + \hat{\zeta}_k^{\ell}(\hat{\delta}_{0,k} - \hat{\delta}_{0,k}^{\ell}) \\ &\triangleq \hat{f}(\hat{\delta}_{0,k}, \hat{\zeta}_k), \end{aligned} \quad (43a)$$

$$\begin{aligned} \hat{\delta}_{1,k}\hat{\xi}_k &\geq \hat{\delta}_{1,k}^{\ell}\hat{\xi}_k^{\ell} + \hat{\delta}_{1,k}^{\ell}(\hat{\xi}_k - \hat{\xi}_k^{\ell}) + \hat{\xi}_k^{\ell}(\hat{\delta}_{1,k} - \hat{\delta}_{1,k}^{\ell}) \\ &\triangleq \hat{f}(\hat{\delta}_{1,k}, \hat{\xi}_k). \end{aligned} \quad (43b)$$

Finally, the optimization problem for IRS phase shifts is given by

$$\mathbf{P}_{3.4} : \max_{\mathbf{E}, \hat{\gamma}, \hat{\delta}_0, \hat{\delta}_1, \hat{\kappa}, \hat{\zeta}, \hat{\xi}} a_0 \sum_{k=1}^K \log_2(1 + \hat{\delta}_{0,k}) + \alpha \hat{f}(\hat{\gamma}, \hat{\delta}_{1,k}) - a_0 \hat{f}(\hat{\kappa}_k) - \alpha \hat{f}(\hat{\gamma}, \hat{\kappa}_k) \quad (44a)$$

$$\text{s.t. (42b) - (42d), (42f), (42h),} \quad (44b)$$

$$\text{Tr}(\mathbf{E}\mathbf{G}_k\mathbf{W}_k\mathbf{G}_k^H) \geq \hat{f}(\hat{\delta}_{0,k}, \hat{\zeta}_k), \forall k, \quad (44c)$$

$$\text{Tr}(\mathbf{E}\mathbf{G}_k\mathbf{W}_k\mathbf{G}_k^H) \geq \hat{f}(\hat{\delta}_{1,k}, \hat{\xi}_k), \forall k. \quad (44d)$$

The problem $\mathbf{P}_{3.4}$ is a convex SDP problem. Then, through using Gaussian Randomization [18], the rank one phase shifts vector can be obtained consequently.

C. Overall BCD Algorithm

The overall BCD-based algorithm for solving the problem $\mathbf{P}_1$ is summarized in **Algorithm 1**. Based on a given sensing time τ , $\mathbf{w}_k$ and Ψ are alternately optimized. The overall algorithm converges until the error tolerance is reached. To tackle the non-convex terms, the SCA and SDR methods are exploited. Besides, the solution of $\mathbf{P}_{2.3}$ serves as a lower bound for that of $\mathbf{P}_2$, the solution of $\mathbf{P}_{3.4}$ serves as a lower bound for that of $\mathbf{P}_3$. Then, the suboptimal solution of the beamforming and the phase shifts can be respectively obtained by performing the eigenvalue decomposition on $\mathbf{W}_k$ and Gaussian Randomization on $\mathbf{E}$, and the optimal sensing time can be obtained by using the exhaustive search. According to [23], the computational complexity for solving $\mathbf{P}_{2.3}$ is $\mathcal{O}(\log(1/\varepsilon_{SCA})((6K + N + 1)L^3 + (6K + N + 1)^2L^2 + (6K + N + 1)^3))$, for solving $\mathbf{P}_{3.4}$ is $\mathcal{O}(\log(1/\varepsilon_{SCA})((5K + M + N + 2)(M + 1)^3 + (5K + M + N + 2)^2(M + 1)^2 + (5K + M + N + 2)^3))$.

IV. SECRECY RATE MAXIMIZATION UNDER IMPERFECT CSI

In this section, we consider a more practical case that the CSI of the primary links and the wiretap links are not perfectly known at the SBS. A joint sensing and transmission security optimization problem is studied under the bounded CSI errors to maximize the sum secrecy rate of SUs by optimizing the beamforming of the SBS and the phase shifts of IRS.

TABLE I
THE BCD-BASED ALGORITHM

Algorithm 1: The BCD-Based Algorithm for $\mathbf{P}_1$

```

for  $\tau = 0 : T$ 
1: Initialize  $\mathbf{W}_k^{(1)}$ ,  $\mathbf{E}^{(1)}$  and auxiliary variables;
2: Set the iteration index  $i = 1$  and the error tolerance  $\varepsilon_{BCD} \rightarrow 0$ ;
3: repeat
4:   Set iteration index  $\ell = 1$  and error tolerance  $\varepsilon_{SCA} \rightarrow 0$ ;
5:   repeat
6:     Solve the problem  $\mathbf{P}_{2.3}$  for given  $\mathbf{E}^{(i)}$ ,  $\mathbf{W}_k^{\ell} = \mathbf{W}_k^{(i)}$ ;
7:     Obtain the solution to the problem,  $\mathbf{W}_k^{\ell}$ ;
8:     Set  $\mathbf{W}_k^{\ell+1} = \mathbf{W}_k^{\ell}$  and  $\ell = \ell + 1$ ;
9:   until  $\left| \frac{\sum_{k=1}^K R_k^{sec}(\mathbf{W}_k^{\ell}, \hat{\delta}_{0,k}^{\ell}, \hat{\delta}_{1,k}^{\ell}, \hat{\kappa}_k^{\ell})}{\sum_{k=1}^K R_k^{sec}(\mathbf{W}_k^{\ell-1}, \hat{\delta}_{0,k}^{\ell-1}, \hat{\delta}_{1,k}^{\ell-1}, \hat{\kappa}_k^{\ell-1})} - 1 \right| \leq \varepsilon_{SCA}$ ;
10:  Set  $\mathbf{W}_k^{(i+1)} = \mathbf{W}_k^{\ell}$ ;
11:  Set iteration index  $\ell = 1$ ;
12:  repeat
13:    Solve the problem  $\mathbf{P}_{3.4}$  for given  $\mathbf{W}_k^{(i)}$ ,  $\mathbf{E}^{\ell} = \mathbf{E}^{(i)}$ ;
14:    Obtain  $\mathbf{E}^{\ell}$ ;
15:    Set  $\mathbf{E}^{\ell+1} = \mathbf{E}^{\ell}$  and  $\ell = \ell + 1$ ;
16:  until  $\left| \frac{\sum_{k=1}^K R_k^{sec}(\mathbf{E}^{\ell}, \hat{\gamma}^{\ell}, \hat{\delta}_{0,k}^{\ell}, \hat{\delta}_{1,k}^{\ell}, \hat{\kappa}_k^{\ell})}{\sum_{k=1}^K R_k^{sec}(\mathbf{E}^{\ell-1}, \hat{\gamma}^{\ell-1}, \hat{\delta}_{0,k}^{\ell-1}, \hat{\delta}_{1,k}^{\ell-1}, \hat{\kappa}_k^{\ell-1})} - 1 \right| \leq \varepsilon_{SCA}$ ;
17:  Set  $\mathbf{E}^{(i+1)} = \mathbf{E}^{\ell}$  and recover  $\Psi^{(i+1)}$  with Gaussian Randomization;
17:  Set  $i = i + 1$ ;
19: until  $\left| \frac{\sum_{k=1}^K R_k^{sec}(\mathbf{W}_k^{(i)}, \Psi^{(i)})}{\sum_{k=1}^K R_k^{sec}(\mathbf{W}_k^{(i-1)}, \Psi^{(i-1)})} - 1 \right| \leq \varepsilon_{BCD}$ ;
20: Store  $\eta(\tau) \triangleq [\mathbf{W}_k^{(i)}, \Psi^{(i)}]$  and  $R(\tau) \triangleq \sum_{k=1}^K r_{s,k}(\tau, \mathbf{W}_k^{(i)}, \Psi^{(i)})$ ;
end
21: Obtain  $\tau^{opt} = \arg \max_{0 \leq \tau \leq T} R$  and  $[\mathbf{W}_k^*, \Psi^*] = \eta(\tau^{opt})$ .
```

A. Bounded CSI Error Model

In practice, the CSI of the primary links cannot be perfectly known at SBS due to the lack of cooperation between the primary network and the secondary network. Recall that $\mathbf{H}_s = \begin{bmatrix} \text{diag}(\mathbf{h}_r^H) \mathbf{H} \\ \mathbf{h}_d^H \end{bmatrix}$ and $\mathbf{V}_n = \begin{bmatrix} \text{diag}(\mathbf{I}_{r,n}^H) \mathbf{G} \\ \mathbf{I}_{d,n}^H \end{bmatrix}$, we consider the worst-case bounded CSI error model [21] to estimate the compositive channel matrices $\mathbf{H}_s$ and $\mathbf{V}_n$, which can be respectively given by

$$\begin{aligned} \mathbf{H}_s &= \bar{\mathbf{H}}_s + \Delta \mathbf{H}_s, \\ \mathcal{H}_s &\triangleq \left\{ \Delta \mathbf{H}_s \in \mathbb{C}^{(M+1) \times L} : \Delta \mathbf{H}_s^H \Delta \mathbf{H}_s \leq \xi_s^2 \right\}, \end{aligned} \quad (45a)$$

$$\begin{aligned} \mathbf{V}_n &= \bar{\mathbf{V}}_n + \Delta \mathbf{V}_n, \\ \mathcal{I}_n &\triangleq \left\{ \Delta \mathbf{V}_n \in \mathbb{C}^{(M+1) \times L} : \Delta \mathbf{V}_n^H \Delta \mathbf{V}_n \leq \xi_{I,n}^2 \right\}, \forall n, \end{aligned} \quad (45b)$$

where $\bar{\mathbf{H}}_s$ and $\bar{\mathbf{V}}_n$ are the estimations at the SBS, $\Delta \mathbf{H}_s$ and $\Delta \mathbf{V}_n$ denote the channel estimation errors, $\mathcal{H}_s$ and $\mathcal{I}_n$ denote the uncertainty regions that bound the uncertain CSI, and ξ_s^2 and $\xi_{I,n}^2$ are the radius of the corresponding uncertainty regions. Moreover, the CSI of wiretap links are generally not perfectly known at SBS since the Eves try to intercept the confidential information of SU-RXs and do not interact with the SBS actively. Note that $\mathbf{F}_j = \begin{bmatrix} \text{diag}(\mathbf{f}_{r,j}^H) \mathbf{G} \\ \mathbf{f}_{d,j}^H \end{bmatrix}$, the bounded CSI error model for the compositive wiretap channel matrix $\mathbf{F}_j$ is given by

$$\begin{aligned} \mathbf{F}_j &= \bar{\mathbf{F}}_j + \Delta \mathbf{F}_j, \\ \mathcal{F}_j &\triangleq \left\{ \Delta \mathbf{F}_j \in \mathbb{C}^{(M+1) \times L} : \Delta \mathbf{F}_j^H \Delta \mathbf{F}_j \leq \xi_{F,j}^2 \right\}, \forall j, \end{aligned} \quad (46)$$

where $\bar{\mathbf{F}}_j$ is the channel estimation, $\Delta \mathbf{F}_j$ denotes the estimation error, $\mathcal{F}_j$ denotes the uncertainty region and $\xi_{F,j}^2$ is the radius of the corresponding uncertainty region.

Since the imperfect CSI is considered, the worst-case secrecy rate maximization problem is formulated by jointly optimizing the beamforming of the SBS and the IRS phase shifts. Similar to the methods adopted for solving the problem $\mathbf{P}_1$, we introduce the auxiliary variables $\hat{z}$, $\mu_{0,k}$, $\mu_{1,k}$ and v_k , and have the following problem.

$$\begin{aligned}
 \mathbf{P}_4 : \quad & \max_{\mathbf{E}, \mathbf{W}_k, \hat{z}, \mu_{0,k}, \mu_{1,k}, v} \\
 & a_0 \sum_{k=1}^K \log_2(1 + \mu_{0,k}) + \alpha g(\hat{z}) \sum_{k=1}^K \log_2(1 + \mu_{1,k}) \\
 & \quad - (a_0 + \alpha g(\hat{z})) \sum_{k=1}^K \log_2(1 + v_k) \quad (47a) \\
 \text{s.t.} \quad & (20d), (20e), (32e) - (32g) \quad (47b) \\
 & g(\hat{z}) \leq 1 - \bar{P}_d, \quad (47c) \\
 & (a_0 + \alpha g(\hat{z})) \sum_{k=1}^K \text{Tr}(\mathbf{W}_k) \leq P_T, \quad (47d) \\
 & \gamma_{0,k} \geq \mu_{0,k}, \forall k, \quad (47e) \\
 & \gamma_{1,k} \geq \mu_{1,k}, \forall k, \quad (47f) \\
 & \max_{\Delta \mathbf{V}_n} \alpha g(\hat{z}) \sum_{k=1}^K \text{Tr}(\mathbf{E} \mathbf{V}_n \mathbf{W}_k \mathbf{V}_n^H) \leq \Gamma_n, \forall n, \quad (47g) \\
 & \max_{\Delta \mathbf{F}_j} \text{Tr}(\mathbf{E} \mathbf{F}_j \mathbf{W}_k \mathbf{F}_j^H) \leq \sigma_{e,j}^2 v_k, \forall k, j, \quad (47h) \\
 & \min_{\Delta \mathbf{H}_s} \frac{L\sigma_s^2 + \text{Tr}(\mathbf{E} \mathbf{H}_s \mathbf{H}_s^H) P - \epsilon}{(\sigma_s^2 / \sqrt{\tau} f_s) \sqrt{2 \text{Tr}(\mathbf{E} \mathbf{H}_s \mathbf{H}_s^H) P / \sigma_s^2 + L}} \geq \sqrt{\hat{z}}, \quad (47i)
 \end{aligned}$$

where $g(\hat{z}) \triangleq \frac{1}{12} e^{-\frac{1}{2} \hat{z}} + \frac{1}{4} e^{-\frac{2}{3} \hat{z}}$ and (47i) is the introduced auxiliary constraint. It is challenging to solve the problem $\mathbf{P}_4$ due to the semi-infinite constraints (47g)-(47i) imposed by the uncertainty of imperfect CSI. To make the constraints tractable, the $\mathcal{S}$ -procedure is adopted to transform the infinite inequality constraints in (47g)-(47i) into equivalent linear matrix inequalities (LMIs). Then, the BCD-based iterative algorithm can be adopted to tackle the non-convex problem with coupled variables.

B. Transformation of the Semi-Infinite Constraint

Considering the bounded CSI error model, the interference power constraint (47g) can be rewritten as

$$\begin{aligned}
 & \alpha g(\hat{z}) \sum_{k=1}^K \text{Tr}(\mathbf{E} \mathbf{V}_n \mathbf{W}_k \mathbf{V}_n^H) \leq \Gamma_n \\
 & \Leftrightarrow \text{Tr} \left(\mathbf{E} \bar{\mathbf{V}}_n \sum_k \mathbf{W}_k \bar{\mathbf{V}}_n^H + \mathbf{E} \Delta \mathbf{V}_n \sum_k \mathbf{W}_k \bar{\mathbf{V}}_n^H \right) \\
 & \quad + \text{Tr} \left(\mathbf{E} \bar{\mathbf{V}}_n \sum_k \mathbf{W}_k \Delta \mathbf{V}_n^H + \mathbf{E} \Delta \mathbf{V}_n \sum_k \mathbf{W}_k \Delta \mathbf{V}_n^H \right) \\
 & \leq \frac{\Gamma_n}{\alpha g(\hat{y})}. \quad (48)
 \end{aligned}$$

Note that $\text{Tr}(\mathbf{ABCD}) = (\text{vec}^H(\mathbf{D}^H)) (\mathbf{C}^H \otimes \mathbf{A}) \text{vec}(\mathbf{B})$, we rewrite the term $\text{Tr}(\mathbf{E} \Delta \mathbf{V}_n \sum_k \mathbf{W}_k \Delta \mathbf{V}_n^H)$ in (48) as

$$\begin{aligned}
 & \text{Tr} \left(\mathbf{E} \Delta \mathbf{V}_n \sum_k \mathbf{W}_k \Delta \mathbf{V}_n^H \right) \\
 & = \text{vec}^H(\Delta \mathbf{V}_n) \left(\sum_k \mathbf{W}_k \otimes \mathbf{E} \right) \text{vec}(\Delta \mathbf{V}_n). \quad (49)
 \end{aligned}$$

Note that $\mathbf{E} = \mathbf{E}^H$ and $\text{Tr}(\mathbf{A}^H \mathbf{B}) = \text{vec}^H(\mathbf{A}) \text{vec}(\mathbf{B})$, the term $\text{Tr}(\mathbf{E} \Delta \mathbf{V}_n \sum_k \mathbf{W}_k \bar{\mathbf{V}}_n^H + \mathbf{E} \bar{\mathbf{V}}_n \sum_k \mathbf{W}_k \Delta \mathbf{V}_n^H)$ can be rewritten as

$$\begin{aligned}
 & \text{Tr} \left(\mathbf{E} \Delta \mathbf{V}_n \sum_k \mathbf{W}_k \bar{\mathbf{V}}_n^H + \mathbf{E} \bar{\mathbf{V}}_n \sum_k \mathbf{W}_k \Delta \mathbf{V}_n^H \right) \\
 & = 2\Re \left\{ \text{Tr} \left(\mathbf{E} \bar{\mathbf{V}}_n \sum_k \mathbf{W}_k \Delta \mathbf{V}_n^H \right) \right\} \\
 & = 2\Re \left\{ \text{vec}^H \left(\sum_k \mathbf{W}_k \bar{\mathbf{V}}_n^H \mathbf{E} \right) \text{vec}(\Delta \mathbf{V}_n^H) \right\}. \quad (50)
 \end{aligned}$$

Thus, the constraint (48) can be transformed into

$$\begin{aligned}
 & \text{vec}^H(\Delta \mathbf{V}_n) \left(\sum_k \mathbf{W}_k \otimes \mathbf{E} \right) \text{vec}(\Delta \mathbf{V}_n) \\
 & \quad + 2\Re \left\{ \text{vec}^H \left(\sum_k \mathbf{W}_k \bar{\mathbf{V}}_n^H \mathbf{E} \right) \text{vec}(\Delta \mathbf{V}_n^H) \right\} \\
 & \quad + \text{Tr} \left(\mathbf{E} \bar{\mathbf{V}}_n \sum_k \mathbf{W}_k \bar{\mathbf{V}}_n^H \right) \leq \frac{\Gamma_n}{\alpha g(\hat{z})}. \quad (51)
 \end{aligned}$$

Besides, the CSI error constraint in (45b) can be rewritten as

$$\text{vec}^H(\Delta \mathbf{V}_n) \text{vec}(\Delta \mathbf{V}_n) \leq \xi_{I,n}^2. \quad (52)$$

Then, we introduce the $\mathcal{S}$ -procedure to tackle the semi-infinite constraint (51). The details of the $\mathcal{S}$ -procedure is given as follows.

Lemma 1: ($\mathcal{S}$ -Procedure [47]): Let a function $f_m(\mathbf{x}) = \mathbf{x}^H \mathbf{A}_m \mathbf{x} + 2\Re\{\mathbf{b}_m^H \mathbf{x}\} + c_m$, $m \in \{1, 2\}$, where $\mathbf{x} \in \mathbb{C}^{N \times 1}$, $\mathbf{A}_m \in \mathbb{H}^N$, $\mathbf{b}_m \in \mathbb{C}^{N \times 1}$, and $c_m \in \mathbb{R}$. Then, the implication $f_1(\mathbf{x}) \leq 0 \Rightarrow f_2(\mathbf{x}) \leq 0$ holds if and only if there exists a $\rho \geq 0$ such that

$$\rho \begin{bmatrix} \mathbf{A}_1 & \mathbf{b}_1 \\ \mathbf{b}_1^H & c_1 \end{bmatrix} - \begin{bmatrix} \mathbf{A}_2 & \mathbf{b}_2 \\ \mathbf{b}_2^H & c_2 \end{bmatrix} \succeq \mathbf{0}, \quad (53)$$

provided that there exists a vector $\hat{\mathbf{x}}$ such that $f_m(\hat{\mathbf{x}}) < 0$.

By applying Lemma 1, the constraint (51) can be rewritten as (54), which is shown at the bottom of next page. Similarly, by applying the transformations and Lemma 1, the semi-infinite constraint (47h) can be equivalently rewritten as (55), which is shown at the bottom of next page.

As for (47i), we introduce the auxiliary variable ϑ and transform it into

$$\frac{L\sigma_s^2 + \vartheta P - \epsilon}{(\sigma_s^2 / \sqrt{\tau} f_s) \sqrt{2\vartheta P / \sigma_s^2 + L}} \geq \sqrt{\hat{z}}, \quad (56a)$$

$$\min_{\Delta \mathbf{H}_s} \text{Tr}(\mathbf{E} \mathbf{H}_s \mathbf{H}_s^H) \geq \vartheta. \quad (56b)$$

It is easy to prove that the left-hand side of (56a) is a monotone increasing function with respect to ϑ . Thus, (56a) and (56b) can ensure that (47i) is satisfied without loss of accuracy. Then, the semi-infinite constraint (56b) can be rewritten as

$$\begin{aligned} \text{Tr}(\mathbf{E}\mathbf{H}_s\mathbf{H}_s^H) &\geq \vartheta \\ \Leftrightarrow \text{Tr}(\mathbf{E}(\bar{\mathbf{H}}_s + \Delta\mathbf{H}_s)(\bar{\mathbf{H}}_s + \Delta\mathbf{H}_s)^H) &\geq \vartheta \\ \Leftrightarrow \text{Tr}(\mathbf{E}\bar{\mathbf{H}}_s\bar{\mathbf{H}}_s^H + \mathbf{E}\Delta\mathbf{H}_s\bar{\mathbf{H}}_s^H + \mathbf{E}\bar{\mathbf{H}}_s\Delta\mathbf{H}_s^H + \mathbf{E}\Delta\mathbf{H}_s\Delta\mathbf{H}_s^H) &\geq \vartheta. \end{aligned} \quad (57)$$

By performing the transformations adopted for solving the constraint (47g), we rewrite (57) as

$$\begin{aligned} &\text{vec}^H(\Delta\mathbf{H}_s)(\mathbf{I}_L \otimes \mathbf{E})\text{vec}(\Delta\mathbf{H}_s) \\ &+ 2\Re\left\{\text{vec}^H(\bar{\mathbf{H}}_s^H\mathbf{E})\text{vec}(\Delta\mathbf{H}_s^H)\right\} \\ &+ \text{Tr}(\mathbf{E}\bar{\mathbf{H}}_s\bar{\mathbf{H}}_s^H) \geq \vartheta. \end{aligned} \quad (58)$$

Then, the CSI error constraint in (45a) can be rewritten as

$$\text{vec}^H(\Delta\mathbf{H}_s)\text{vec}(\Delta\mathbf{H}_s) \leq \xi_s^2. \quad (59)$$

Thus, by applying Lemma 1, (58) can be rewritten as (60), which is shown at the bottom of this page.

C. Beamforming Optimization for Given IRS Phase Shifts

By introducing the auxiliary variables $\tilde{\zeta}_k$ and $\tilde{\xi}_k$, the beamforming optimization for given Ψ under the imperfect CSI is given by

$$\begin{aligned} \mathbf{P}_{4.1} : \quad &\max_{\mathbf{W}_k, \hat{z}, \vartheta, \mu_0, \mu_1, v, \tilde{\zeta}, \tilde{\xi}} \\ &a_0 \sum_{k=1}^K \log_2(1 + \mu_{0,k}) + \alpha g(\hat{z}) \sum_{k=1}^K \log_2(1 + \mu_{1,k}) \\ &- (a_0 + \alpha g(\hat{z})) \sum_{k=1}^K \log_2(1 + v_k) \end{aligned} \quad (61a)$$

$$\text{s.t. (54), (55), (56a), (60),} \quad (61b)$$

$$(a_0 + \alpha g(\hat{z})) \sum_{k=1}^K \text{Tr}(\mathbf{W}_k) \leq P_T, \quad (61c)$$

$$\text{Tr}(\mathbf{W}_k \mathbf{G}_k^H \mathbf{E} \mathbf{G}_k) \geq \hat{f}(\mu_{0,k}, \tilde{\zeta}_k), \forall k, \quad (61d)$$

$$\sum_{j \neq k}^K \text{Tr}(\mathbf{W}_j \mathbf{G}_k^H \mathbf{E} \mathbf{G}_k) + \sigma_{r,k}^2 \leq \tilde{\zeta}_k, \forall k, \quad (61e)$$

$$\text{Tr}(\mathbf{W}_k \mathbf{G}_k^H \mathbf{E} \mathbf{G}_k) \geq \hat{f}(\mu_{1,k}, \tilde{\xi}_k), \forall k, \quad (61f)$$

$$\sum_{j \neq k}^K \text{Tr}(\mathbf{W}_j \mathbf{G}_k^H \mathbf{E} \mathbf{G}_k) + P|z_k|^2 + \sigma_{r,k}^2 \leq \tilde{\xi}_k, \forall k, \quad (61g)$$

$$\text{Rank}(\mathbf{W}_k) \leq 1, \forall k, \quad (61h)$$

$$\mathbf{W}_k \succeq \mathbf{0}, \forall k, \quad (61i)$$

where $\hat{f}(\mu_{0,k}, \tilde{\zeta}_k)$ and $\hat{f}(\mu_{1,k}, \tilde{\xi}_k)$ are the first-order Taylor expansion of $\mu_{0,k}\tilde{\zeta}_k$ and $\mu_{1,k}\tilde{\xi}_k$, respectively, given as

$$\hat{f}(\mu_{0,k}, \tilde{\zeta}_k) \triangleq \mu_{0,k}^{\ell} \tilde{\zeta}_k^{\ell} + \mu_{0,k}^{\ell} (\tilde{\zeta}_k - \tilde{\zeta}_k^{\ell}) + \tilde{\zeta}_k^{\ell} (\mu_{0,k} - \mu_{0,k}^{\ell}), \quad (62a)$$

$$\hat{f}(\mu_{1,k}, \tilde{\xi}_k) \triangleq \mu_{1,k}^{\ell} \tilde{\xi}_k^{\ell} + \mu_{1,k}^{\ell} (\tilde{\xi}_k - \tilde{\xi}_k^{\ell}) + \tilde{\xi}_k^{\ell} (\mu_{1,k} - \mu_{1,k}^{\ell}). \quad (62b)$$

The problem $\mathbf{P}_{4.1}$ is still non-convex due to the objective function and the constraints (54), (56a), (61c) and (61h). By using the SCA, the lower bound of the terms $f(\hat{z}, \mu_{1,k}) \triangleq g(\hat{y}) \sum_{k=1}^K \log_2(1 + \hat{\delta}_{1,k})$ and $f(\hat{z}, v_k) \triangleq (a_0 + \alpha g(\hat{z})) \sum_{k=1}^K \log_2(1 + v_k)$ in the objective function can be respectively given by

$$\begin{aligned} f(\hat{z}, \mu_{1,k}) &\geq f(\hat{z}^{\ell}, \mu_{1,k}^{\ell}) + \sum_{k=1}^K \nabla_{\mu_{1,k}} f(\hat{z}^{\ell}, \mu_{1,k}^{\ell}) (\mu_{1,k} - \mu_{1,k}^{\ell}) \\ &+ \nabla_{\hat{z}} f(\hat{z}^{\ell}, \mu_{1,k}^{\ell}) (\hat{z} - \hat{z}^{\ell}) \triangleq \hat{f}(\hat{z}, \mu_{1,k}), \end{aligned} \quad (63a)$$

$$\begin{aligned} f(\hat{z}, v_k) &\geq f(\hat{z}^{\ell}, v_k^{\ell}) + \sum_{k=1}^K \nabla_{v_k} f(\hat{z}^{\ell}, v_k^{\ell}) (v_k - v_k^{\ell}) \\ &+ \nabla_{\hat{z}} f(\hat{z}^{\ell}, v_k^{\ell}) (\hat{z} - \hat{z}^{\ell}) \triangleq \hat{f}(\hat{z}, v_k). \end{aligned} \quad (63b)$$

Similarly, the lower bound of $f(\hat{z}, \mathbf{W}_k) \triangleq g(\hat{z}) \sum_{k=1}^K \text{Tr}(\mathbf{W}_k)$ at the ℓ th iteration of the SCA for a given feasible

$$\begin{bmatrix} \iota \mathbf{I}_{(M+1) \times L} - \sum_k \mathbf{W}_k \otimes \mathbf{E} & -\text{vec}\left(\sum_k \mathbf{W}_k \bar{\mathbf{V}}_n^H \mathbf{E}\right) \\ -\text{vec}^H\left(\sum_k \mathbf{W}_k \bar{\mathbf{V}}_n^H \mathbf{E}\right) & -\iota \xi_{I,n}^2 - \text{Tr}\left(\mathbf{E} \bar{\mathbf{V}}_n \sum_k \mathbf{W}_k \bar{\mathbf{V}}_n^H\right) + \frac{\Gamma_n}{\alpha g(\hat{z})} \end{bmatrix} \succeq \mathbf{0}, \forall n. \quad (54)$$

$$\begin{bmatrix} \rho \mathbf{I}_{(M+1) \times L} - \sum_k \mathbf{W}_k \otimes \mathbf{E} & -\text{vec}\left(\sum_k \mathbf{W}_k \bar{\mathbf{F}}_j^H \mathbf{E}\right) \\ -\text{vec}^H\left(\sum_k \mathbf{W}_k \bar{\mathbf{F}}_j^H \mathbf{E}\right) & -\iota \xi_{F,j}^2 - \text{Tr}\left(\mathbf{E} \bar{\mathbf{F}}_j \sum_k \mathbf{W}_k \bar{\mathbf{F}}_j^H\right) + \sigma_{e,j}^2 v_k \end{bmatrix} \succeq \mathbf{0}, \forall k, j. \quad (55)$$

$$\begin{bmatrix} \lambda \mathbf{I}_{(M+1) \times L} + \mathbf{I}_L \otimes \mathbf{E} & \text{vec}\left(\bar{\mathbf{H}}_s^H \mathbf{E}\right) \\ \text{vec}^H\left(\bar{\mathbf{H}}_s^H \mathbf{E}\right) & -\lambda \xi_s^2 + \text{Tr}\left(\mathbf{E} \bar{\mathbf{H}}_s \bar{\mathbf{H}}_s^H\right) - \vartheta \end{bmatrix} \succeq \mathbf{0}. \quad (60)$$

point $(\mathbf{W}_k^\ell, \hat{z}^\ell)$ can be given by

$$\begin{aligned} \hat{f}(\hat{z}, \mathbf{W}_k) &\triangleq \\ &f(\hat{z}^\ell, \mathbf{W}_k^\ell) + \sum_{k=1}^K \text{Tr} \left(\nabla_{\mathbf{W}_k} f(\hat{z}^\ell, \mathbf{W}_k^\ell)^H (\mathbf{W}_k - \mathbf{W}_k^\ell) \right) \\ &+ \nabla_{\hat{z}} f(\hat{z}^\ell, \mathbf{W}_k^\ell)(\hat{z} - \hat{z}^\ell). \end{aligned} \quad (64)$$

Note that the constraint (54) is not guaranteed to be convex with respect to $\hat{z}$. Thus, the term $f(\hat{z}) \triangleq \frac{\Gamma_n}{3\alpha g(\hat{z})}$ is transformed into its linear form by using the first-order Taylor expansion, denoted by $\hat{f}(\hat{z})$. As for the constraint (61c), the upper bound of $f(\hat{z}, \vartheta) \triangleq \sqrt{2\vartheta P/\sigma_s^2 + L}\sqrt{\hat{z}}$ at the ℓ th iteration of the SCA for a given feasible point $(\vartheta^\ell, \hat{z}^\ell)$ is given by $\hat{f}(\hat{z}, \vartheta) \triangleq f(\hat{z}^\ell, \vartheta^\ell) + \nabla_{\vartheta} f(\hat{z}^\ell, \vartheta^\ell)(\vartheta - \vartheta^\ell) + \nabla_{\hat{z}} f(\hat{z}^\ell, \vartheta^\ell)(\hat{z} - \hat{z}^\ell)$. Then, by applying the SDR to relax (61h), the problem $\mathbf{P}_{4.1}$ can be rewritten as (65a)–(65e), which is shown at the bottom of the next page. The problem $\mathbf{P}_{4.2}$ is convex. Then, the eigenvalue decomposition or Gaussian Randomization can be adopted to obtain the suboptimal solution of the beamforming.

D. Phase Shifts Optimization for Given Beamforming

For any given $\mathbf{W}_k$, the phase shifts optimization problem is formulated as

$$\begin{aligned} \mathbf{P}_{4.3} : \quad &\max_{\mathbf{E}, \hat{z}, \vartheta, \mu_0, \mu_1, v, \tilde{\zeta}, \tilde{\xi}} \\ &a_0 \sum_{k=1}^K \log_2(1 + \mu_{0,k}) + \alpha g(\hat{z}) \sum_{k=1}^K \log_2(1 + \mu_{1,k}) \\ &- (a_0 + \alpha g(\hat{z})) \sum_{k=1}^K \log_2(1 + v_k) \quad (66a) \\ \text{s.t.} \quad &(47c), (47d), (55), (60), (65d), (65e), \quad (66b) \\ &\text{Tr}(\mathbf{E} \mathbf{G}_k \mathbf{W}_k \mathbf{G}_k^H) \geq \mu_{0,k}, \forall k, \quad (66c) \\ &\text{Tr}(\mathbf{E} \mathbf{G}_k \mathbf{W}_k \mathbf{G}_k^H) \geq \mu_{1,k}, \forall k, \quad (66d) \\ &\text{Rank}(\mathbf{E}) = 1, \quad (66e) \\ &\mathbf{E} \succeq \mathbf{0}, [\mathbf{E}]_{m,m} \leq 1, \forall m. \quad (66f) \end{aligned}$$

The non-convexity in the problem $\mathbf{P}_{4.3}$ originates from the objective function and the constraints (66c)–(66e). Then, by introducing the auxiliary variables $\tilde{\zeta}_k$ and $\tilde{\xi}_k$, and applying the SCA and SDR methods, the problem $\mathbf{P}_{4.3}$ can be reformulated as

$$\begin{aligned} \mathbf{P}_{4.4} : \quad &\max_{\mathbf{E}, \hat{z}, \vartheta, \mu_0, \mu_1, v, \tilde{\zeta}, \tilde{\xi}} a_0 \sum_{k=1}^K \log_2(1 + \mu_{0,k}) \\ &+ \alpha \hat{f}(\mu_{1,k}, \hat{z}) - \hat{f}(\hat{z}, v_k) \quad (67a) \\ \text{s.t.} \quad &(66b), (66f), \quad (67b) \end{aligned}$$

$$\text{Tr}(\mathbf{E} \mathbf{G}_k \mathbf{W}_k \mathbf{G}_k^H) \geq \hat{f}(\mu_{0,k}, \tilde{\zeta}_k), \forall k, \quad (67c)$$

$$\sum_{j \neq k}^K \text{Tr}(\mathbf{E} \mathbf{G}_k \mathbf{W}_j \mathbf{G}_k^H) + \sigma_{r,k}^2 \leq \tilde{\zeta}_k, \forall k, \quad (67d)$$

$$\text{Tr}(\mathbf{E} \mathbf{G}_k \mathbf{W}_k \mathbf{G}_k^H) \geq \hat{f}(\mu_{1,k}, \tilde{\xi}_k), \forall k, \quad (67e)$$

$$\begin{aligned} &\sum_{j \neq k}^K \text{Tr}(\mathbf{E} \mathbf{G}_k \mathbf{W}_j \mathbf{G}_k^H) \\ &+ P \text{Tr}(\mathbf{E} \mathbf{R}_k \mathbf{R}_k^H) + \sigma_{r,k}^2 \leq \tilde{\xi}_k, \forall k, \quad (67f) \end{aligned}$$

where $\hat{f}(\mu_{1,k}, \hat{z})$, $\hat{f}(\hat{z}, v_k)$, $\hat{f}(\mu_{0,k}, \tilde{\zeta}_k)$ and $\hat{f}(\mu_{1,k}, \tilde{\xi}_k)$ are the first-order Taylor expansion of the term $f(\mu_{1,k}, \hat{z}) \triangleq g(\hat{z}) \sum_{k=1}^K \log_2(1 + \mu_{1,k})$, $f(\hat{z}, v_k) \triangleq (a_0 + \alpha g(\hat{z})) \sum_{k=1}^K \log_2(1 + v_k)$, $\mu_{0,k} \tilde{\zeta}_k$ and $\mu_{1,k} \tilde{\xi}_k$, respectively. Finally, problem $\mathbf{P}_{4.4}$ is a convex SDP problem, and the suboptimal solution of the phase shifts matrix can be obtained by applying Gaussian Randomization. Similarly, the computational complexity for solving $\mathbf{P}_{4.2}$ is $\mathcal{O}(\log(1/\varepsilon_{SCA})((KJ + 5K + N + 1)L^3 + (KJ + 5K + N + 1)^2 L^2 + (KJ + 5K + N + 1)^3 + 3))$, for solving $\mathbf{P}_{4.4}$ is $\mathcal{O}(\log(1/\varepsilon_{SCA})((KJ + 4K + M + N + 2)(M + 1)^3 + (KJ + 4K + M + N + 2)^2(M + 1)^2 + (KJ + 4K + M + N + 2)^3 + 3))$.

V. SIMULATION RESULTS

In this section, simulation results are provided to evaluate the performance of our proposed schemes. Based on [5], [23], and [26], the simulation parameters are set as follows. The error tolerance is set as $\varepsilon_{BCD} = \varepsilon_{SCA} = 0.01$. The frame duration is set as $T = 100$ ms. The sampling frequency is set as $f_s = 6$ MHz. The predefined probability of detection is set as $\bar{P}_d = 0.9$. The target probability of false alarm is set as $\bar{P}_f = 0.01$. The number of IRS elements is set as $M = 10$. The number of SBS antennas is set as $N = 4$. The inactive probability of the PU-TX is set as $\Pr(\mathcal{H}_0) = 0.8$. The transmit power of PU-TX is set as $P = -10$ dBm. The variance of the noise is set as -110 dBm. The variance of the noise is set as -10 dBm. The maximum tolerable interference is set as $\Gamma_l = -5$ dB. The maximum transmit power of the SBS is set as $P_T = 10$ dBm. Besides, the channels are modeled as $\mathbf{h}_\beta = \sqrt{\beta_0(d)^{-v_\beta}} \mathbf{g}_\beta$, where β_0 is the reference path loss, d is the link distance, v_β is the path loss exponent and $\mathbf{g}_\beta \sim \mathcal{CN}(0, \mathbf{I})$ is the Rayleigh fading component, respectively, where $\beta \in \{D, SI, R\}$ is the set of the SBS-user link, SBS-IRS link and IRS-user link. The path loss exponents are set as $v_D = 3.5$, $v_{SI} = 2.2$ and $v_R = 2.2$, respectively. The locations of the PU-TX, SBS and IRS are set as $(5, 0, 20)$, $(5, 100, 20)$ and $(0, 50, 2)$, respectively. Then, the locations of the SU-RXs are set as $(5, 60, 0)$ and $(5, 75, 0)$, respectively. The locations of the PU-RXs are set as $(25, 0, 0)$ and $(45, 0, 0)$, respectively. The locations of the Eves are set as $(5, 70, 0)$ and $(5, 80, 0)$, respectively.

Our proposed scheme under the perfect CSI case is marked as ‘Proposed, perfect CSI’, while that under the imperfect CSI case is marked as ‘Proposed, imperfect CSI’. Three baseline schemes are considered. The first baseline scheme is that the IRS phase shifts are generated randomly, and beamforming is optimized to maximize the sum average secrecy rate subject to (14) and (15), marked as ‘Random phase shifts’. The second

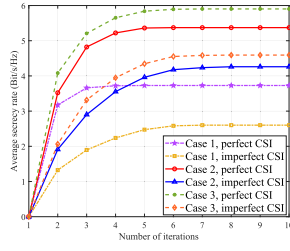


Fig. 2. The convergence of the proposed BCD-based algorithm.

baseline scheme is that the IRS only assists the secure transmission does not improve the spectrum sensing, the detection threshold is set to ensure target probability of detection and the probability of false alarm under this scheme is given by $\tilde{P}_f = Q\left(\sqrt{\frac{2\gamma+1}{N}}Q^{-1}(\bar{P}_d) + \sqrt{\frac{\tau f_s}{N}}\gamma\right)$, and beamforming and phase shifts are optimized for the maximization of the secrecy rate subject to (14) and (15), which is marked by ‘No IRS-enhanced SS’. The third baseline scheme is the conventional spectrum sensing-based CRN without the assistance of IRS, and the beamforming is optimized for the maximization of the secrecy rate subject to the transmit power constraint and the interference power constraints, which is marked as ‘No IRS’.

Fig. 2 shows the convergence of the proposed BCD-based algorithm under the perfect and imperfect CSI for different numbers of SBS antennas N , IRS elements M and SU-RX K . In particular, three cases are considered: Case 1 with $N = 2$, $M = 5$ and $K = 2$; Case 2 with $N = 4$, $M = 10$ and $K = 2$; Case 3 with $N = 4$, $M = 10$ and $K = 4$. It is observed that the proposed BCD-based algorithm under the perfect and imperfect CSI can converge to a stationary point within 10 iterations, which verified the effectiveness of our proposed algorithm. The secrecy rate achieved under the imperfect CSI is lower than that achieved under the perfect CSI due to the uncertainty of the imperfect CSI. Then, Compared to Case 1, Case 2 needs more iterations for convergence. The reason is that the solution space is enlarged with the increase of N and M . It is also observed that Case 3 needs more iterations for convergence than Case 2, the fact is that the solution space is enlarged with the increase of K . Moreover, the performance

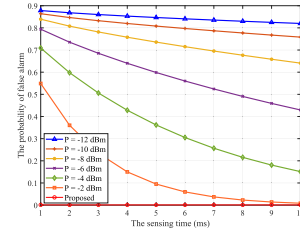


Fig. 3. Probability of false alarm versus the sensing time.

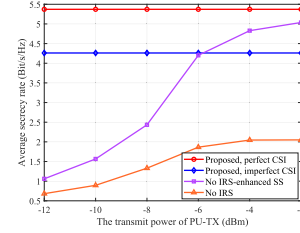


Fig. 4. Secrecy rate versus the transmit power of PU-TX.

achieved under the Case 3 is better than Case 2, this is because a higher system diversity gain can be obtained by increasing the number of SU-RXs due to the exploitation of multi-user diversity technologies. Furthermore, the reason that Case 2 is better than Case 1 is that more degrees of freedom (DoFs) are provided and a higher active and passive beamforming gain can be achieved with the increase of N and M , respectively.

Fig. 3 shows the probability of false alarm versus the sensing time for different transmit power levels of the PU-TX. It is observed that the probability of false in our proposed scheme is the lowest and remains constant while that in other schemes decreases with the increase of the sensing time. The reason is that, in the proposed scheme, the probability of false alarm can be fixed to a low value by adjusting the detection threshold while the target probability of detection can be guaranteed with the assistance of IRS. However, in other schemes, the detection threshold must be set to ensure the target probability of detection, which leads to a high probability of false alarm in the low SNR. It is concluded that our proposed scheme can significantly improve the sensing accuracy.

Fig. 4 illustrates the sum secrecy rate versus the transmit power of the PU-TX. We observe that our proposed

$$\mathbf{P}_{4.2} : \max_{\mathbf{W}_k, \hat{\mathbf{z}}, \vartheta, \mu_0, \mu_1, \mathbf{v}, \tilde{\xi}} a_0 \sum_{k=1}^K \log_2(1 + \mu_{0,k}) + \alpha \hat{f}(\mu_{1,k}, \hat{\mathbf{z}}) - \hat{f}(\hat{\mathbf{z}}, \hat{\mathbf{v}}_k) \quad (65a)$$

$$\text{s.t. (47c), (55), (60), (61d) - (61h),} \quad (65b)$$

$$a_0 \sum_{k=1}^K \text{Tr}(\mathbf{W}_k) + \alpha \hat{f}(\hat{\mathbf{z}}, \mathbf{W}_k) \leq P_T, \quad (65c)$$

$$\begin{bmatrix} \mathbf{I}_{(M+1) \times L} - \sum_k \mathbf{W}_k \otimes \mathbf{E} & -\text{vec}\left(\sum_k \mathbf{W}_k \bar{\mathbf{V}}_n^H \mathbf{E}\right) \\ -\text{vec}^H\left(\sum_k \mathbf{W}_k \bar{\mathbf{V}}_n^H \mathbf{E}\right) & -\iota \xi_{L,n}^2 - \text{Tr}\left(\mathbf{E} \bar{\mathbf{V}}_n \sum_k \mathbf{W}_k \bar{\mathbf{V}}_n^H\right) + \hat{f}(\hat{\mathbf{z}}) \end{bmatrix} \succeq \mathbf{0}, \forall n, \quad (65d)$$

$$\frac{\sigma_s^2}{\tau f_s} \hat{f}(\vartheta, \hat{\mathbf{z}}) - L \sigma_s^2 - \vartheta P + \epsilon \leq 0. \quad (65e)$$

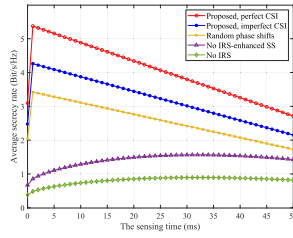


Fig. 5. Secrecy rate versus the sensing time.

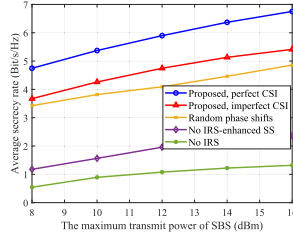


Fig. 6. Secrecy rate versus the maximum transmit power of the SBS.

design excels in the low SNR regime, achieving high secrecy performance. The key factor behind this observation lies in the accuracy of the spectrum sensing process, which directly influences the achievable secrecy performance in opportunistic access CRNs. Our proposed design ensures not only a high probability of detection but also maintains a low and constant probability of false alarm. In contrast, other schemes exhibit poor secrecy performance in the low SNR regime, and their false alarm probability varies with SNR, often reaching high levels under such conditions. This leads us to conclude that our proposed design is highly effective in low SNR scenarios, offering both consistently high secrecy rates and low false alarm probabilities.

Fig. 5 presents the sum secrecy rate versus the sensing time. We observe a tradeoff between sensing and transmission in the conventional No IRS-enhanced SS and No IRS cases. Our proposed schemes outperform the baseline schemes, achieving the highest value at 1 ms. The reason is twofold. First, the proposed detection threshold and IRS-enhanced spectrum sensing maintain fixed values for sensing results, i.e., P_d and P_f , irrespective of the sensing time. Second, as the sensing time increases, the instantaneous secrecy rate decreases due to reduced available transmission time. These findings indicate that our proposed schemes offer enhanced sensing precision in shorter time frames, leading to increased transmission time and improved secrecy performance in CRNs. This eliminates the need for a tradeoff between sensing and transmission.

Fig. 6 shows the sum secrecy rate versus the maximum transmit power of the SBS. It is observed that the secrecy rate increases monotonically with the transmit power of the SBS and that under the perfect CSI is the highest among all schemes. Besides, the secrecy rate achieved under the imperfect CSI is lower than that achieved under the perfect CSI due to the uncertainty of the imperfect CSI, but is still better compared with the baseline schemes. This indicates that the exploitation of IRS is beneficial for improving the communication security in CRNs and the formulated joint sensing and transmission security optimization is very effective. Moreover, it is observed that the performance achieved

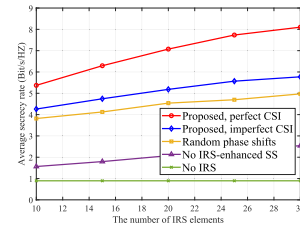


Fig. 7. Secrecy rate versus the number of IRS elements.

by ‘No IRS-enhanced SS’ is lower than our proposed scheme. The reason is that the proposed IRS-enhanced spectrum sensing approach can increase the sensing performance, thereby increasing the secrecy rate, even without the adjustment of phase shifts. It is concluded that exploiting IRS in both the spectrum sensing and secure transmission can further enhance the communication security of CRNs.

Fig. 7 shows the sum secrecy rate versus the number of IRS elements. It is observed that the secrecy performance achieved in our proposed schemes increases with the number of IRS elements. The fact is that the increase of IRS elements provides more DoFs, thereby resulting in a higher passive beamforming gain, which further increases the transmission rate at the SU-RXs and decreases that at the Eves. In contrast, the performance improvement of the baseline schemes is limited since that the phase shifts are not optimized.

VI. CONCLUSION

In this paper, an IRS-enhanced secure opportunistic access CRN was studied under both the perfect and imperfect CSI cases. The IRS was exploited to enhance both the sensing accuracy and the secrecy performance. Moreover, a new approach was proposed to ensure the target probability of detection by exploiting IRS-enhanced spectrum sensing. A joint sensing and transmission security optimization problem was formulated to maximize the sum secrecy rate of SUs. A safe approximation was adopted to transform the probability of detection into a tractable expression. For the perfect CSI case, an efficient BCD-based algorithm was exploited to optimize the beamforming and the phase shifts alternately. Then, for the imperfect CSI case, several transformations and the $\mathcal{S}$ -procedure were adopted to tackle the infinite non-convex constraints. Simulation results demonstrated that the proposed BCD-based algorithm is very efficient. Furthermore, the proposed IRS-enhanced spectrum sensing ensures both the required probability of detection and the probability of false alarm within significantly reduced sensing time, eliminating the need to consider a tradeoff between sensing and transmission. In conclusion, our proposed schemes substantially enhance sensing accuracy and secrecy performance simultaneously in low SNR scenarios compared to the benchmark schemes.

REFERENCES

- [1] Q. Wu et al., “A unified cognitive learning framework for adapting to dynamic environments and tasks,” *IEEE Wireless Commun.*, vol. 28, no. 6, pp. 208–216, Dec. 2021.
- [2] M. Amjad, M. H. Rehmani, and S. Mao, “Wireless multimedia cognitive radio networks: A comprehensive survey,” *IEEE Commun. Surveys Tuts.*, vol. 20, no. 2, pp. 1056–1103, 2nd Quart., 2018.

- [3] G. Wu, F. Zhou, G. Ding, Q. Wu, and X.-Y. Li, "An efficient heterogeneous edge-cloud learning framework for spectrum data compression," *IEEE Trans. Mobile Comput.*, vol. 22, no. 7, pp. 3823–3839, Jul. 2023.
- [4] F. Zhou, Y. Li, X. Zhang, Q. Wu, X. Lei, and R. Q. Hu, "Cognitive semantic communication systems driven by knowledge graph," in *Proc. IEEE Int. Conf. Commun.*, May 2022, pp. 4860–4865.
- [5] Y.-C. Liang, Y. Zeng, E. C. Y. Peh, and A. T. Hoang, "Sensing-throughput tradeoff for cognitive radio networks," *IEEE Trans. Wireless Commun.*, vol. 7, no. 4, pp. 1326–1337, Apr. 2008.
- [6] S. Choi, H. Park, and T. Hwang, "Optimal beamforming and power allocation for sensing-based spectrum sharing in cognitive radio networks," *IEEE Trans. Veh. Technol.*, vol. 63, no. 1, pp. 412–417, Jan. 2014.
- [7] F. Zhou, N. C. Beaulieu, J. Cheng, Z. Chu, and Y. Wang, "Robust max-min fairness resource allocation in sensing-based wideband cognitive radio with SWIPT: Imperfect channel sensing," *IEEE Syst. J.*, vol. 12, no. 3, pp. 2361–2372, Sep. 2018.
- [8] F. Zhou and N. C. Beaulieu, "An improved and more accurate expression for a PDF related to eigenvalue-based spectrum sensing," *IEEE Syst. J.*, vol. 13, no. 2, pp. 1320–1323, Jun. 2019.
- [9] T. Yucek and H. Arslan, "A survey of spectrum sensing algorithms for cognitive radio applications," *IEEE Commun. Surveys Tuts.*, vol. 11, no. 1, pp. 116–130, 1st Quart., 2009.
- [10] X. Liu, K. Zheng, K. Chi, and Y.-H. Zhu, "Cooperative spectrum sensing optimization in energy-harvesting cognitive radio networks," *IEEE Trans. Wireless Commun.*, vol. 19, no. 11, pp. 7663–7676, Nov. 2020.
- [11] L. Li, H. Jiang, and H. He, "Deep transfer cooperative sensing in cognitive radio," *IEEE Wireless Commun. Lett.*, vol. 10, no. 6, pp. 1354–1358, Jun. 2021.
- [12] F. Shen, G. Ding, Z. Wang, and Q. Wu, "UAV-based 3D spectrum sensing in spectrum-heterogeneous networks," *IEEE Trans. Veh. Technol.*, vol. 68, no. 6, pp. 5711–5722, Jun. 2019.
- [13] J. Perez, J. Via, L. Vielva, and D. Ramirez, "Online detection and SNR estimation in cooperative spectrum sensing," *IEEE Trans. Wireless Commun.*, vol. 21, no. 4, pp. 2521–2533, Apr. 2022.
- [14] Z. Shi, W. Gao, S. Zhang, J. Liu, and N. Kato, "Machine learning-enabled cooperative spectrum sensing for non-orthogonal multiple access," *IEEE Trans. Wireless Commun.*, vol. 19, no. 9, pp. 5692–5702, Sep. 2020.
- [15] Y. Wu, R. Schober, D. W. K. Ng, C. Xiao, and G. Caire, "Secure massive MIMO transmission with an active eavesdropper," *IEEE Trans. Inf. Theory*, vol. 62, no. 7, pp. 3880–3900, Jul. 2016.
- [16] T. Zhang, Y. Cai, Y. Huang, T. Q. Duong, and W. Yang, "Secure transmission in cognitive MIMO relaying networks with outdated channel state information," *IEEE Access*, vol. 4, pp. 8212–8224, 2016.
- [17] F. Zhou, Z. Chu, H. Sun, R. Q. Hu, and L. Hanzo, "Artificial noise aided secure cognitive beamforming for cooperative MISO-NOMA using SWIPT," *IEEE J. Sel. Areas Commun.*, vol. 36, no. 4, pp. 918–931, Apr. 2018.
- [18] Q. Wu and R. Zhang, "Intelligent reflecting surface enhanced wireless network via joint active and passive beamforming," *IEEE Trans. Wireless Commun.*, vol. 18, no. 11, pp. 5394–5409, Nov. 2019.
- [19] C. Pan et al., "Reconfigurable intelligent surfaces for 6G systems: Principles, applications, and research directions," *IEEE Commun. Mag.*, vol. 59, no. 6, pp. 14–20, Jun. 2021.
- [20] Q. Wu et al., "Intelligent reflecting surface-aided wireless communications: A tutorial," *IEEE Trans. Commun.*, vol. 69, no. 5, pp. 3313–3351, May 2021.
- [21] J. Yuan, Y.-C. Liang, J. Joung, G. Feng, and E. G. Larsson, "Intelligent reflecting surface-assisted cognitive radio system," *IEEE Trans. Commun.*, vol. 69, no. 1, pp. 675–687, Jan. 2021.
- [22] Y. Wu, F. Zhou, Q. Wu, Y. Huang, and R. Q. Hu, "Resource allocation for IRS-assisted sensing-enhanced wideband CR networks," in *Proc. IEEE Int. Conf. Commun. Workshops (ICC Workshops)*, Jun. 2021, pp. 1–6.
- [23] D. Xu, X. Yu, Y. Sun, D. W. K. Ng, and R. Schober, "Resource allocation for IRS-assisted full-duplex cognitive radio systems," *IEEE Trans. Commun.*, vol. 68, no. 12, pp. 7376–7394, Dec. 2020.
- [24] C. Huang, A. Zappone, G. C. Alexandropoulos, M. Debbah, and C. Yuen, "Reconfigurable intelligent surfaces for energy efficiency in wireless communication," *IEEE Trans. Wireless Commun.*, vol. 18, no. 8, pp. 4157–4170, Aug. 2019.
- [25] L. Zhang, C. Pan, Y. Wang, H. Ren, and K. Wang, "Robust beamforming design for intelligent reflecting surface aided cognitive radio systems with imperfect cascaded CSI," *IEEE Trans. Cognit. Commun. Netw.*, vol. 8, no. 1, pp. 186–201, Mar. 2022.
- [26] Y. Wu, F. Zhou, W. Wu, Q. Wu, R. Q. Hu, and K.-K. Wong, "Multi-objective optimization for spectrum and energy efficiency tradeoff in IRS-assisted CRNs with NOMA," *IEEE Trans. Wireless Commun.*, vol. 21, no. 8, pp. 6627–6642, Aug. 2022.
- [27] Y. Pei, Y.-C. Liang, L. Zhang, K. C. Teh, and K. H. Li, "Secure communication over MISO cognitive radio channels," *IEEE Trans. Wireless Commun.*, vol. 9, no. 4, pp. 1494–1502, Apr. 2010.
- [28] Y. Pei, Y.-C. Liang, K. C. Teh, and K. H. Li, "Secure communication in multiantenna cognitive radio networks with imperfect channel state information," *IEEE Trans. Signal Process.*, vol. 59, no. 4, pp. 1683–1693, Apr. 2011.
- [29] B. Fang, Z. Qian, W. Zhong, and W. Shao, "AN-aided secrecy precoding for SWIPT in cognitive MIMO broadcast channels," *IEEE Commun. Lett.*, vol. 19, no. 9, pp. 1632–1635, Sep. 2015.
- [30] Z. Xiang, W. Yang, G. Pan, Y. Cai, and Y. Song, "Physical layer security in cognitive radio inspired NOMA network," *IEEE J. Sel. Top. Signal Process.*, vol. 13, no. 3, pp. 700–714, Jun. 2019.
- [31] S. Bhattacharjee, "Friendly jamming assisted secure cooperative multicasting in cognitive radio-NOMA networks," in *Proc. IEEE Globecom Workshops*, Waikoloa, HI, USA, Dec. 2019, pp. 1–6.
- [32] W. Wu et al., "IRS-enhanced energy detection for spectrum sensing in cognitive radio networks," *IEEE Wireless Commun. Lett.*, vol. 10, no. 10, pp. 2254–2258, Oct. 2021.
- [33] X. Li, Q. Zhu, and Y. Wang, "IRS-assisted crowd spectrum sensing in B5G cellular IoT networks," in *Proc. Int. Conf. Wireless Commun. Signal Process. (WCSP)*, Oct. 2020, pp. 761–765.
- [34] S. Lin, B. Zheng, F. Chen, and R. Zhang, "Intelligent reflecting surface-aided spectrum sensing for cognitive radio," *IEEE Wireless Commun. Lett.*, vol. 11, no. 5, pp. 928–932, May 2022.
- [35] A. Nasser, H. Al Haj Hassan, A. Mansour, K.-C. Yao, and L. Nuaymi, "Intelligent reflecting surfaces and spectrum sensing for cognitive radio networks," *IEEE Trans. Cognit. Commun. Netw.*, vol. 8, no. 3, pp. 1497–1511, Apr. 2022.
- [36] W. Wu, Z. Wang, F. Zhou, B. Wang, Q. Wu, and N. Al-Dhahir, "Joint sensing and transmission optimization in IRS-assisted CRNs: Throughput maximization," in *Proc. GLOBECOM IEEE Global Commun. Conf.*, Dec. 2022, pp. 2438–2443.
- [37] J. Ge, Y.-C. Liang, S. Li, and Z. Bai, "RIS-enhanced spectrum sensing: How many reflecting elements are required to achieve a detection probability close to 1?" *IEEE Trans. Wireless Commun.*, vol. 21, no. 10, pp. 8600–8615, Oct. 2022.
- [38] X. Yu, D. Xu, and R. Schober, "Enabling secure wireless communications via intelligent reflecting surfaces," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2019, pp. 1–6.
- [39] X. Yu, D. Xu, Y. Sun, D. W. K. Ng, and R. Schober, "Robust and secure wireless communications via intelligent reflecting surfaces," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 11, pp. 2637–2652, Nov. 2020.
- [40] Q. Wang, F. Zhou, R. Q. Hu, and Y. Qian, "Energy efficient robust beamforming and cooperative jamming design for IRS-assisted MISO networks," *IEEE Trans. Wireless Commun.*, vol. 20, no. 4, pp. 2592–2607, Apr. 2021.
- [41] S. Hong, C. Pan, H. Ren, K. Wang, and A. Nallanathan, "Artificial-noise-aided secure MIMO wireless communications via intelligent reflecting surface," *IEEE Trans. Commun.*, vol. 68, no. 12, pp. 7851–7866, Dec. 2020.
- [42] L. Dong, H. M. Wang, and H. Xiao, "Secure cognitive radio communication via intelligent reflecting surface," *IEEE Trans. Commun.*, vol. 69, no. 7, pp. 4678–4690, Jul. 2021.
- [43] L. Dong, H.-M. Wang, H. Xiao, and J. Bai, "Secure intelligent reflecting surface assisted MIMO cognitive radio transmission," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Mar. 2021, pp. 1–6.
- [44] H.-B. Jeon, S.-H. Park, J. Park, K. Huang, and C.-B. Chae, "An energy-efficient aerial backhaul system with reconfigurable intelligent surface," *IEEE Trans. Wireless Commun.*, vol. 21, no. 8, pp. 6478–6494, Aug. 2022.
- [45] X. Wu, J. Ma, Z. Xing, C. Gu, X. Xue, and X. Zeng, "Secure and energy efficient transmission for IRS-assisted cognitive radio networks," *IEEE Trans. Cogn. Commun. Netw.*, vol. 8, no. 1, pp. 170–185, Mar. 2022.
- [46] M. K. Simon, *Probability Distributions Involving Gaussian Random Variables*. Cham, Switzerland: Springer, 2006.
- [47] S. P. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge, U.K.: Cambridge Univ. Press, 2004.
- [48] M. Chiani, D. Dardari, and M. K. Simon, "New exponential bounds and approximations for the computation of error probability in fading channels," *IEEE Trans. Wireless Commun.*, vol. 24, no. 5, pp. 840–845, May 2003.



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